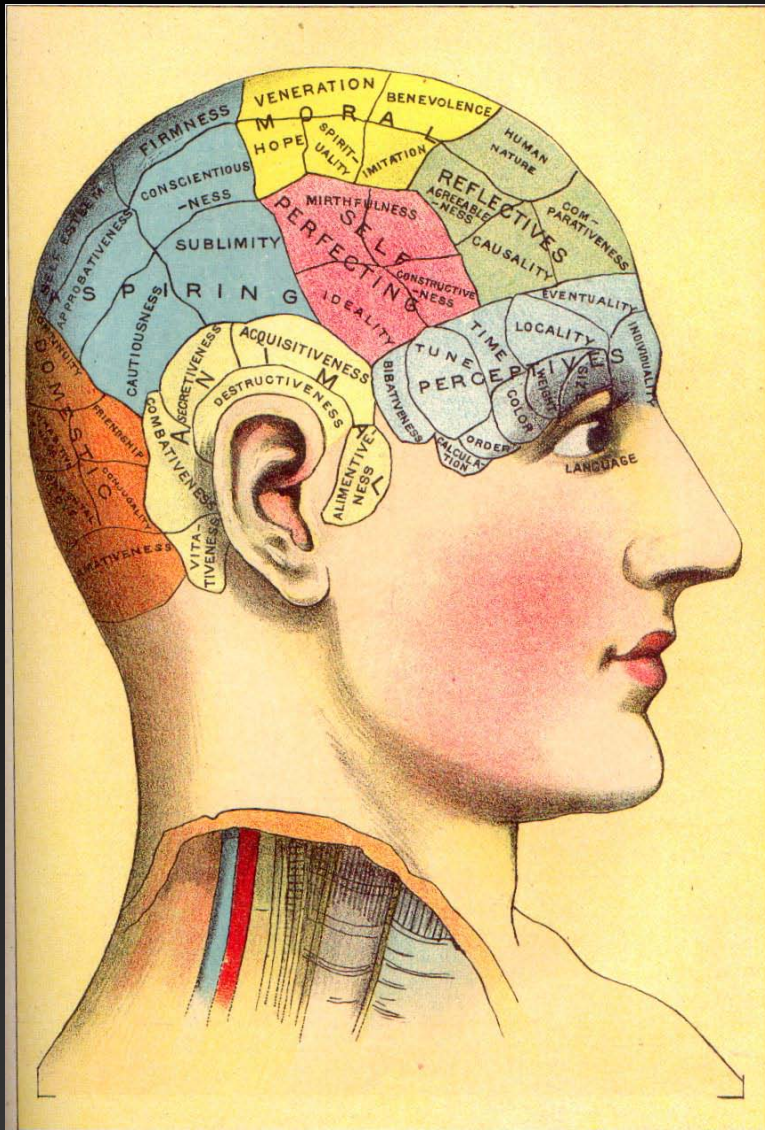




- ✦ Understanding pain better
- ✦ Insurance and medicolegal considerations
- ✦ Treatment options

G & T
Test

2
minutes
hand in
ice

Throbbing	3
Shooting	1
Stabbing	
Sharp	6
Cramping	3
Gnawing	3
Hot-burning	5
Aching	6
Heavy	1
Tender	
Splitting	
Tiring- exhausting	
Sickening	1
Fearful	
Punishing - cruel	4

intensity

0	
1	1
2	2
3	1
4	1
5	1
6	1
7	7
8	
9	

distress

0	
1	
2	4
3	2
4	1
5	
6	1
7	3
8	2
9	

Sensations experienced following stabbings



- punched
- cold
- 'something going in'
- sharp
- hit

The relationship between MRI appearance of the lumbar spine and LBP, age and occupation in males

- ✱ 149 working men, office workers to draymen
- ✱ Disc degen age related commonest at L5/S1
- ✱ LBP commoner with age but not related to degen
- ✱ No relation between occupation and MRI
- ✱ 45% had abnormal MRIs
- ✱ 32% of asymptomatic subjects had abnormal MRI
- ✱ 47% of subjects with LBP had normal MRI

Savage et al Eur Spine J 1997



Relationship between x ray changes and clinical parameters in osteoarthritis of the hip

	Minimal joint space group (mm)			<i>x² trend p <</i>
	>2.5	>1.5-2.5	<1.5	
	(n=136) <i>No (%)</i>	(n=31) <i>No (%)</i>	(n=28) <i>No (%)</i>	
Pain severity >4	65 (48)	17 (55)	18 (64)	0.10
Walk >30 min/day	96 (72)	17 (55)	17 (61)	0.14
Use a walking stick	13 (10)	2 (6)	8 (29)	0.02
Take analgesics	98 (73)	21 (68)	22 (79)	0.65
Use topical treats	62 (46)	16 (52)	16 (57)	0.24
See physiotherapist	11 (8)	1 (3)	3 (11)	0.91
See osteopath/chiro	12 (9)	1 (3)	2 (7)	0.53

Birrell et al. Ann Rheum Dis 2000;59:857-863

Population surveys on chronic pain

- ★ 50.4% of patients in 29 practices in Grampian region reported chronic pain (Elliott et al, Lancet, 1999)
- ★ Older age, female gender, housing tenure & employment status all associated with chronic pain
- ★ 32% of 'new episode' LBP patients in 2 South Manchester practices had repeat consultations within 3 months; after that time, 90% stopped consulting but most still experienced pain & disability in the following year (Croft et al, BMJ, 1998)

Prevalence of whiplash symptoms following placebo collision

- ✱ Complex protocol involving simulated rear end collision
- ✱ 51 subjects, history and physical examination taken at T0, T1 just after, T2 3 days after, T4 4 weeks after.
- ✱ Various psychological measures taken alongside symptom analysis
- ✱ Symptoms (fright, pain, dizziness, headache, nausea, tinnitus, fatigue, vomiting) initially in 20% reduced to 10% by T4
- ✱ A group of psychological factors predicted 83-92% development of symptoms
- ✱ Emotionally unstable, less happy with life and more concerned with health more at risk



Castro et al 2001

Recovery from low back pain

Greenough C Acta Scand (suppl254) 1993; 64

☀ Study

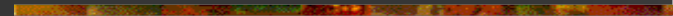
- ☀ 287 patients, back pain, onset incident related
- ☀ 90% follow up, for up to 5 years
- ☀ Measures included: diagnosis, psychometrics, ODQ, injury severity, compensation, time off work

☀ Outcome (multivariate analysis)

- ☀ Outcome affected by: compensation, age, sex time off work social group
 - ☀ Not affected by: diagnosis, injury score, time, neurology
 - ☀ Compensation and psych distress = 35% of variance
- 

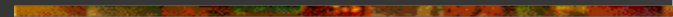
The pursuit of compensation is bad for your health....

- ✱ Various outcome studies have shown worse outcome for similar injuries in those pursuing litigation
- ✱ Persistence and severity of disability not related to severity of accident
- ✱ Interim payments better than lump sum
- ✱ No miraculous cure with settlement

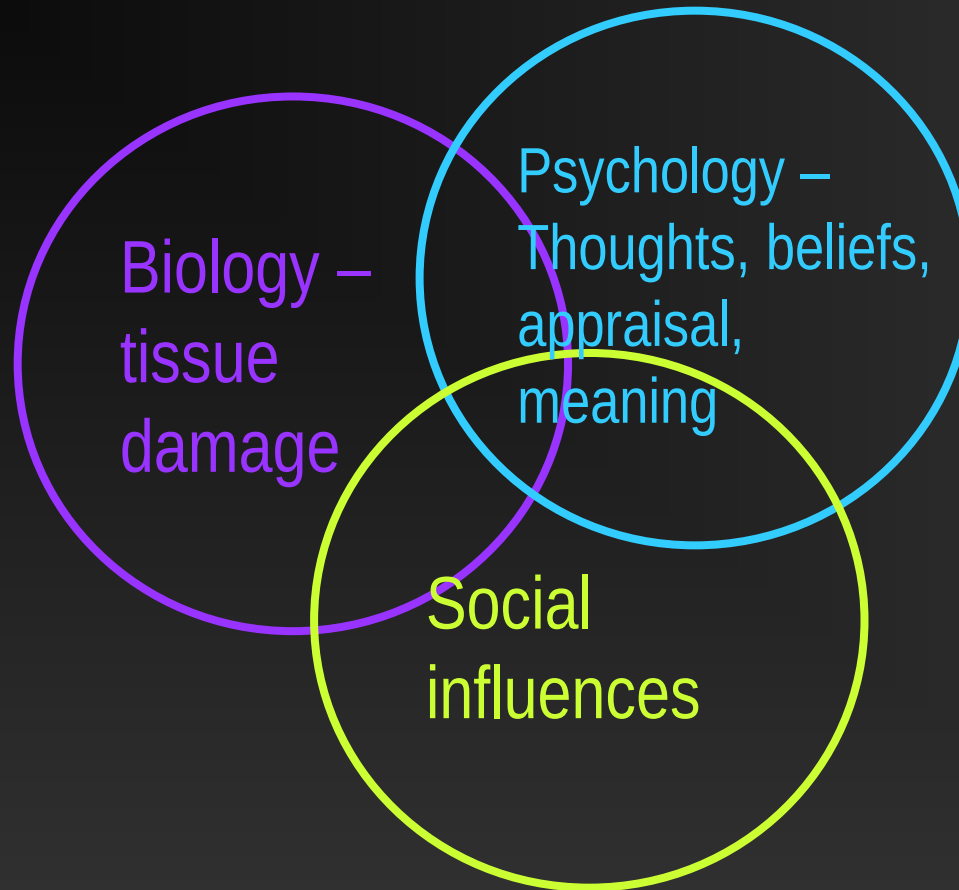


Pain and hypnotic analgesia

- ✱ PET and fMRI demonstrate cortical activity correlates with pain experience in a number of areas of the brain
- ✱ Hypnosis can alter unpleasantness of pain with commensurate changes to activity in the cingulate
- ✱ Activity in sensory SI cortex unchanged
- ✱ Numerous studies showing similar findings

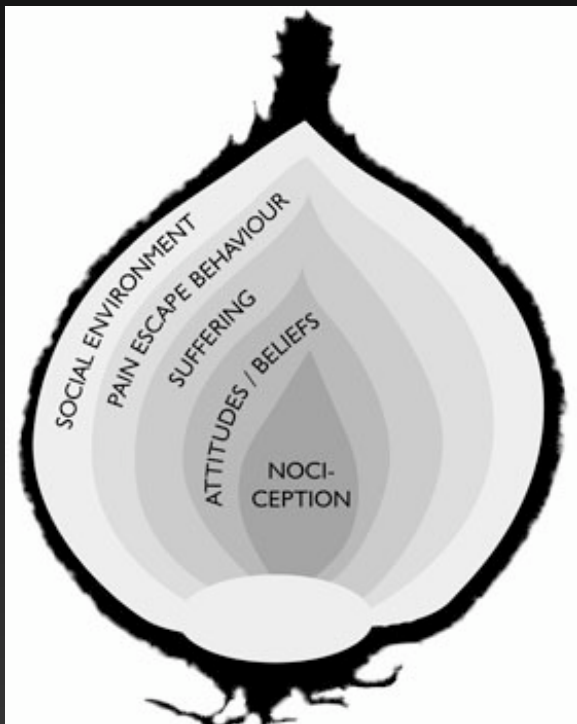


Bio-psycho-social model

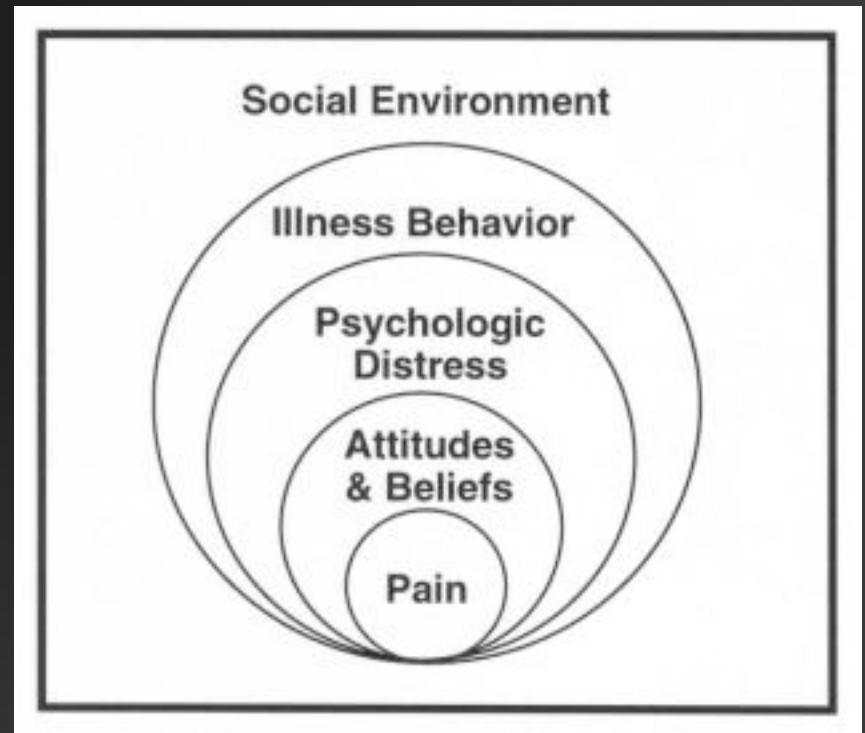


Biopsychosocial model

- Term first coined by Engel in 1950's with concept of 'pain prone patient'
- Modified and developed by many



Loeser



Main & Waddell

Cold pressor test

Hand in iced water for 2 minutes, pain scored on VAS. PET scanning of cortex

Wide variety of subjective experience, despite uniform stimulus

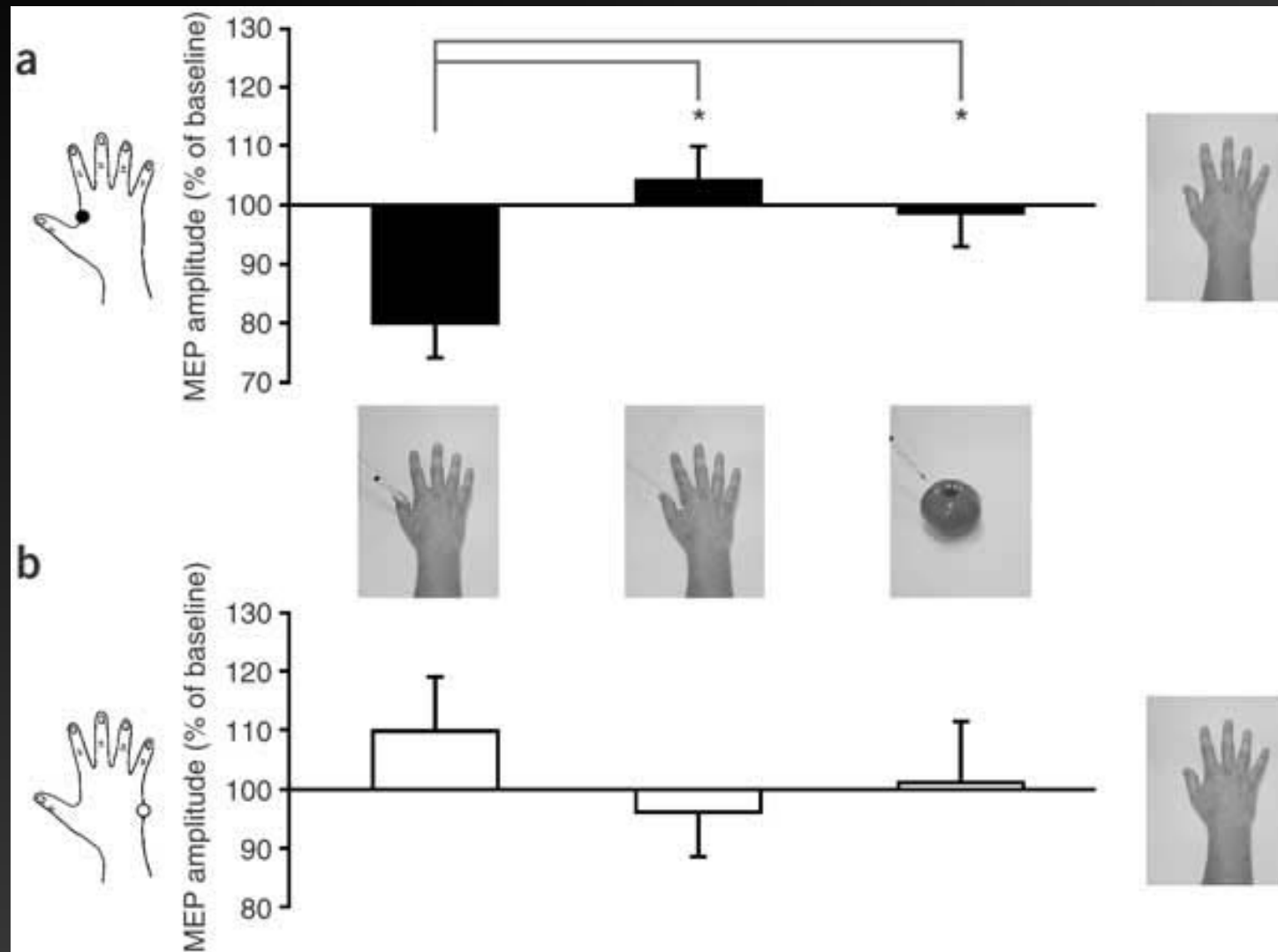
Activity in SI sensory cortex constant

Activity in cingulate related to subjective experience



Coghill et al 2003

Transcranial magnetic stimulation highlights the sensorimotor side of empathy for pain Avenanti et al, Nature neuroscience 2005



Catastrophizing = focus on threat,

- ✱ rumination,
- ✱ magnification,
- ✱ helplessness

“I can feel my neck click when I move”

overestimating threat,

“the bones are crumbling and I’ll be paralysed”

underestimating resources to deal with it

“nobody understands how to fix it, and I just can’t bear any more pain”

Catastrophic thinking enhances attentional interference

Crombez Pain 75 (1998) 187-198

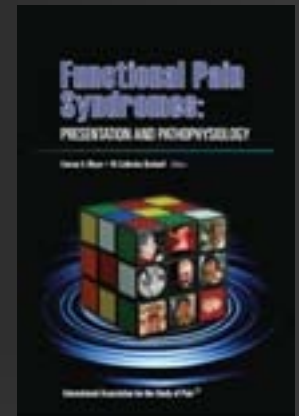
- ✱ student volunteers, electric stimulation, auditory task discrimination
- ✱ measurement of tendency to catastrophise
- ✱ threat induced: expectation of shock or told pain fibre stimulation

conclusions

- ✱ catastrophisers showed marked task interference after threat
 - ✱ amplification of somatosensory information
 - ✱ priming of fear mechanisms

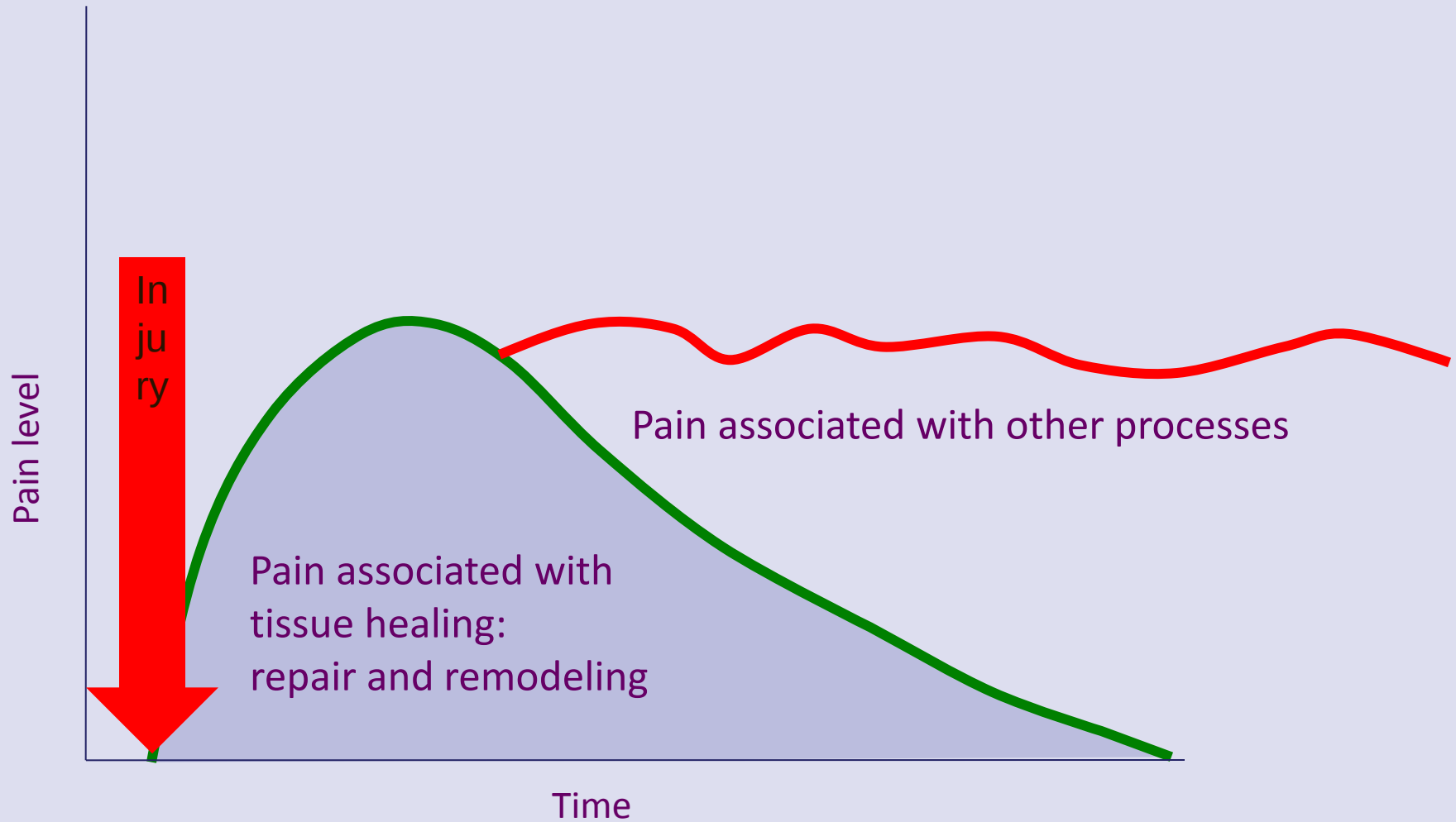
Low back pain

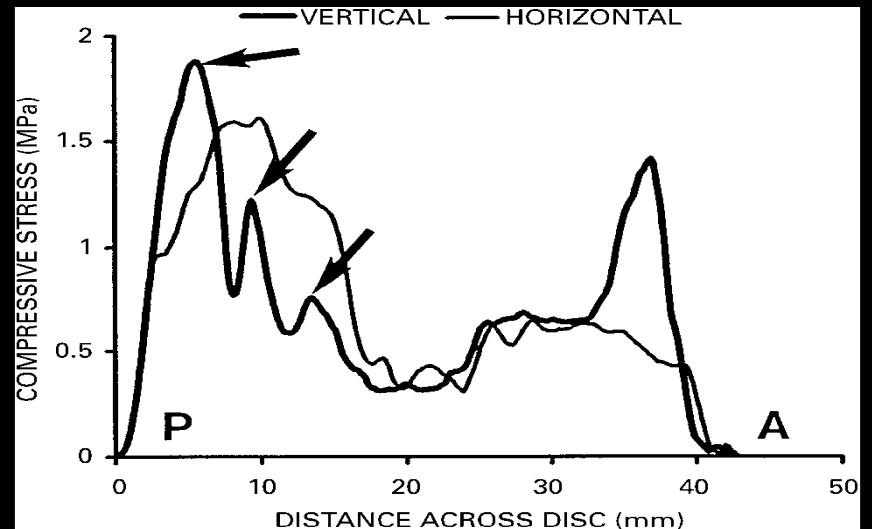
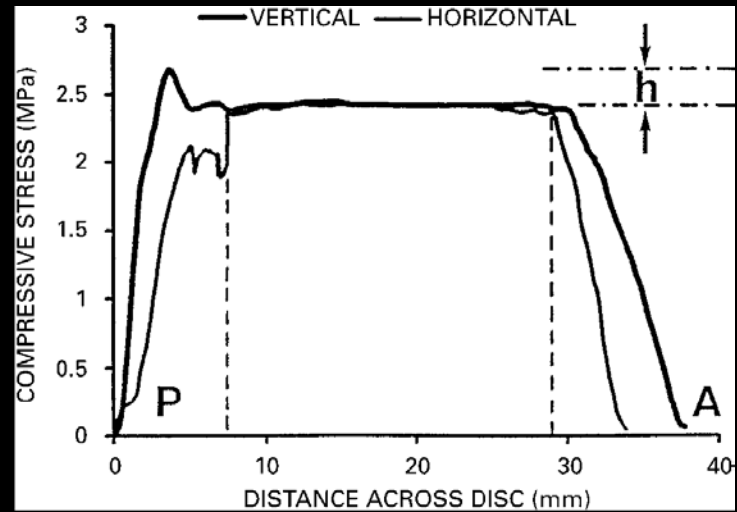
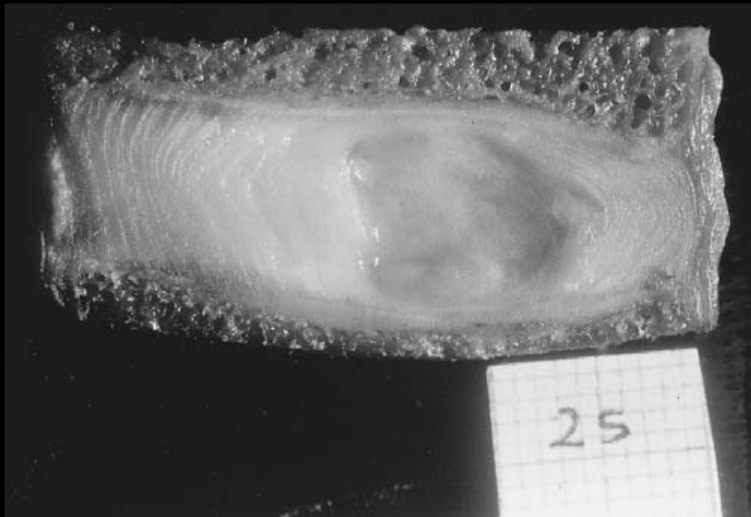
- Mechanisms of low back pain
 - End organ dysfunction
 - Altered nervous system processing
- Domains considered
 - Distinct causative events
 - Definable biologic abnormality
 - Genetics
 - Co-occurrence with other pain syndromes
 - Co-occurrence with emotional dysfunction
 - Evidence of abnormal CNS function



Robinson & Apkarian 2009

Persistent musculoskeletal pain

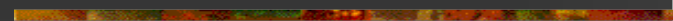


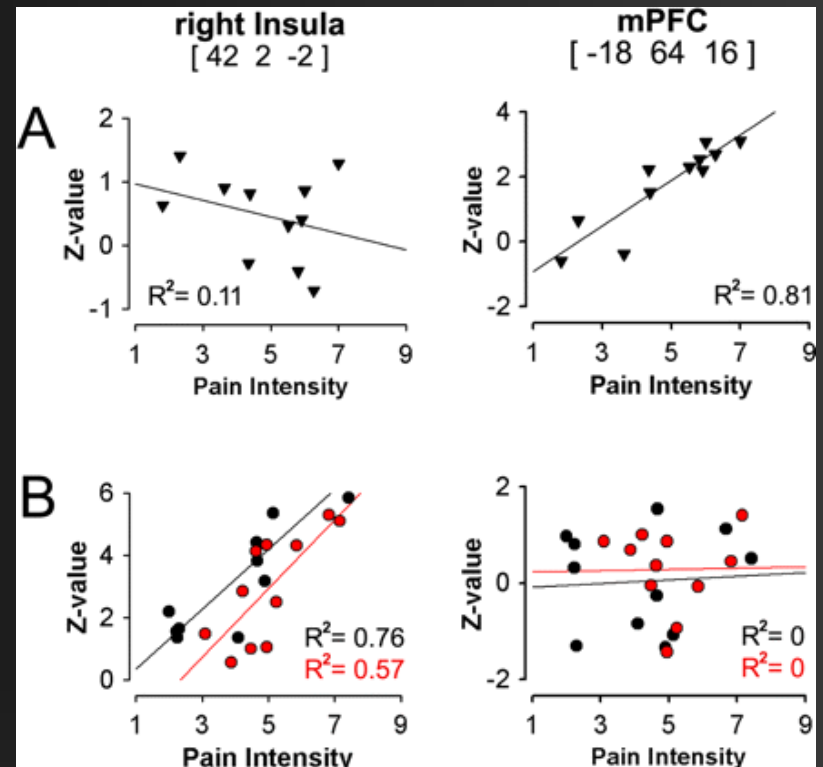
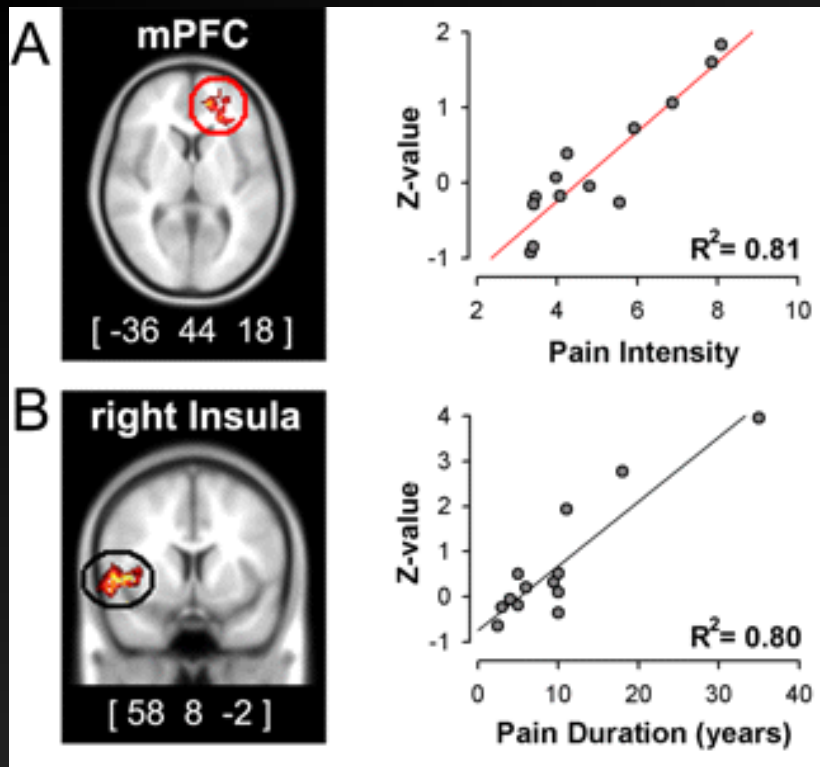


Adams et al 1996

Evidence for abnormal CNS function in LBP

- ✱ Changes to mechanical pain pressure thresholds in LBP subjects - Giesecke 2004
- ✱ Difference between thermal pain responses applied to back in LBP subjects and spontaneous brain activity - Baliki 2006
- ✱ Chronic back pain involves activation of insular, proportional to duration of symptoms
- ✱ Brain activity in LBP subjects differs from controls even at rest - Baliki 2008
- ✱ Anatomical abnormalities – grey matter atrophy (dorsolateral PFC – different from fibromyalgia) – Kuchinad 2007
- ✱ Cognitive abnormalities linked to insula and DLPFC



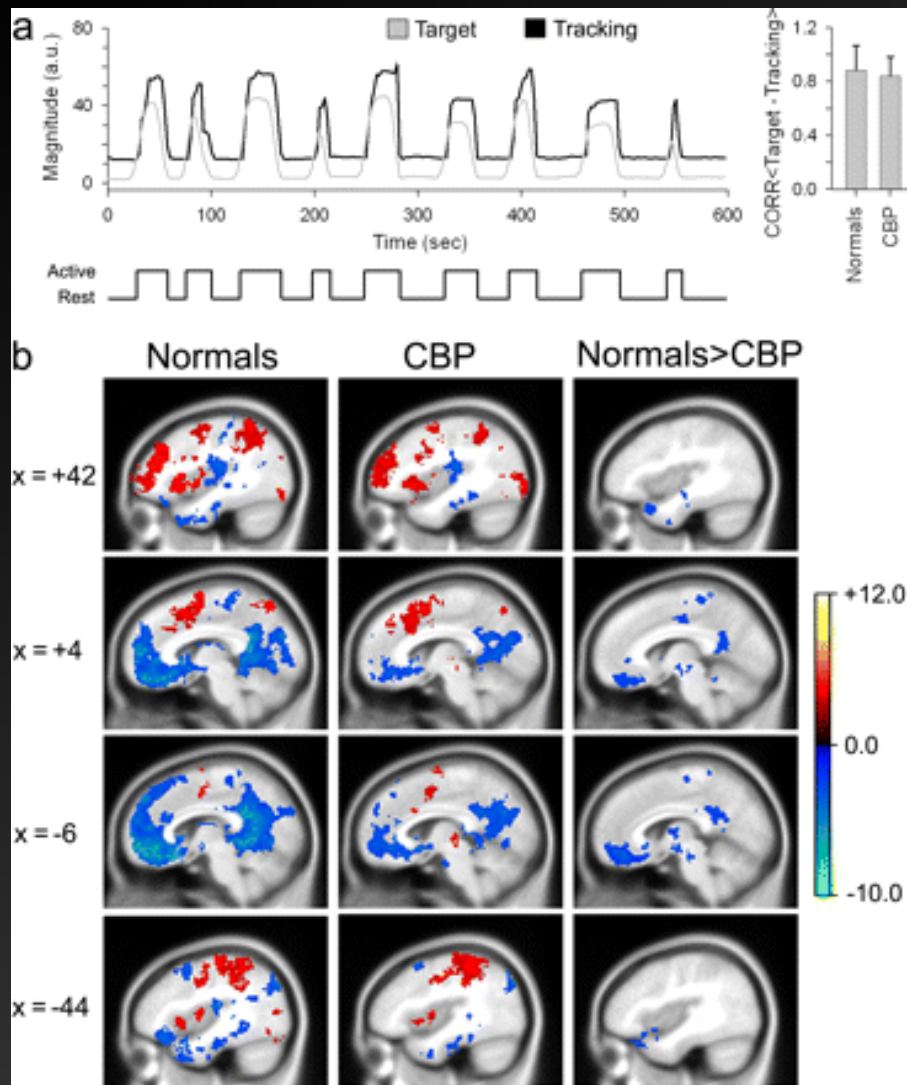


“These findings suggest that subjective spontaneous pain of CBP involves specific spatiotemporal neuronal mechanisms, distinct from those observed for acute experimental pain, implicating a salient role for emotional brain concerning the self.”

Baliki 2006

Evidence for abnormal CNS function in LBP

- ✱ Changes to mechanical pain pressure thresholds in LBP subjects - Giesecke 2004
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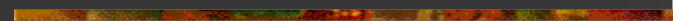


“In conclusion, these findings suggest that the brain of a chronic pain patient is not simply a healthy brain processing pain information, but rather is altered by the persistent pain in a manner reminiscent of other neurological conditions associated with cognitive impairments.”

Baliki 2008

Evidence for abnormal CNS function in LBP

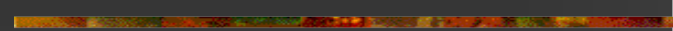
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- ✱ Cognitive abnormalities linked to insula and DLPFC



Pain without nociception

“Recent imaging data display activity of the near entire ‘pain matrix’ without any nociceptive input’.....this is not to say that such pain (sometimes referred to as psychogenic) is any less real than ‘physically’ defined pain; indeed neuroimaging has highlighted the physiological reality of such experiencesrather these studies illustrate how powerful the mind can be in terms of activating specific networks within the brain to produce a realistic and vivid experience.”

Irene Tracey BJA 2008

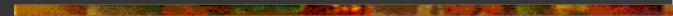


What do we mean by organic?

Do these models really help?

For many the issue remains:
is in the body or is it in the mind?

Most patients, surgeons and pain specialists still think pain is essentially end organ dysfunction.



Somatic / functional pain complaints

- ☀ Numerous conditions associated with persistent pain with unclear pathology
 - ☀ Low back pain
 - ☀ Irritable bowel syndrome
 - ☀ Painful bladder/interstitial cystitis
 - ☀ Burning mouth syndrome
 - ☀ Temporomandibular joint pain TMJ
 - ☀ Vulvodynia
 - ☀ Atypical chest pain
- ☀ Term functional to emphasize alteration of function rather than structure
- ☀ Various common features not explained by end organ pathology

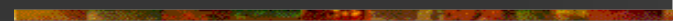
Somatic / functional pain complaints

☀ Problems

- ☀ Lack of pathology leads to psychological formulation or disbelief
- ☀ Complex development of diagnostic categorisation by symptom check lists
- ☀ Power of politics and patient support groups
- ☀ Other somatic complaints often ignored or not recognised
- ☀ Endorgan changes may be secondary to central autonomic changes

☀ Question:

- ☀ Is there one or many syndromes?



Somatic / functional pain complaints

Epidemiology and clinical features

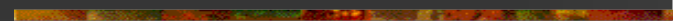
☀ Evidence for common processes at core of functional pain complaints

- ☀ Overlap with normal population
 - Large number with symptom vs small number consulting
 - Commonality of neurocognitive factors: hypervigilance to bodily sensations, catastrophising, symptom related anxiety and illness behaviour capable of being shaped by medical treatment, culture, parental influence etc
- ☀ Demonstrable 'reality' of cortical changes
- ☀ Greater prevalence in women (dominance of neurocognitive mechanisms)
- ☀ History of adverse life events and stress sensitivity
- ☀ Comorbidity with disorders of mood and affect
- ☀ Response to centrally acting therapies (Drug CBT)

Somatic / functional pain complaints

Biological mechanisms

- ✱ Altered pain perception and processing
 - ✱ Heightened sensitivity to a variety of stimuli
 - ✱ Evidence for alteration to brain structure
 - ✱ Evidence for alterations in neurocognitive function
 - ✱ Evidence for increased activity in central arousal circuits and sympathetic nervous system function
 - ✱ Evidence for common genetic susceptibility

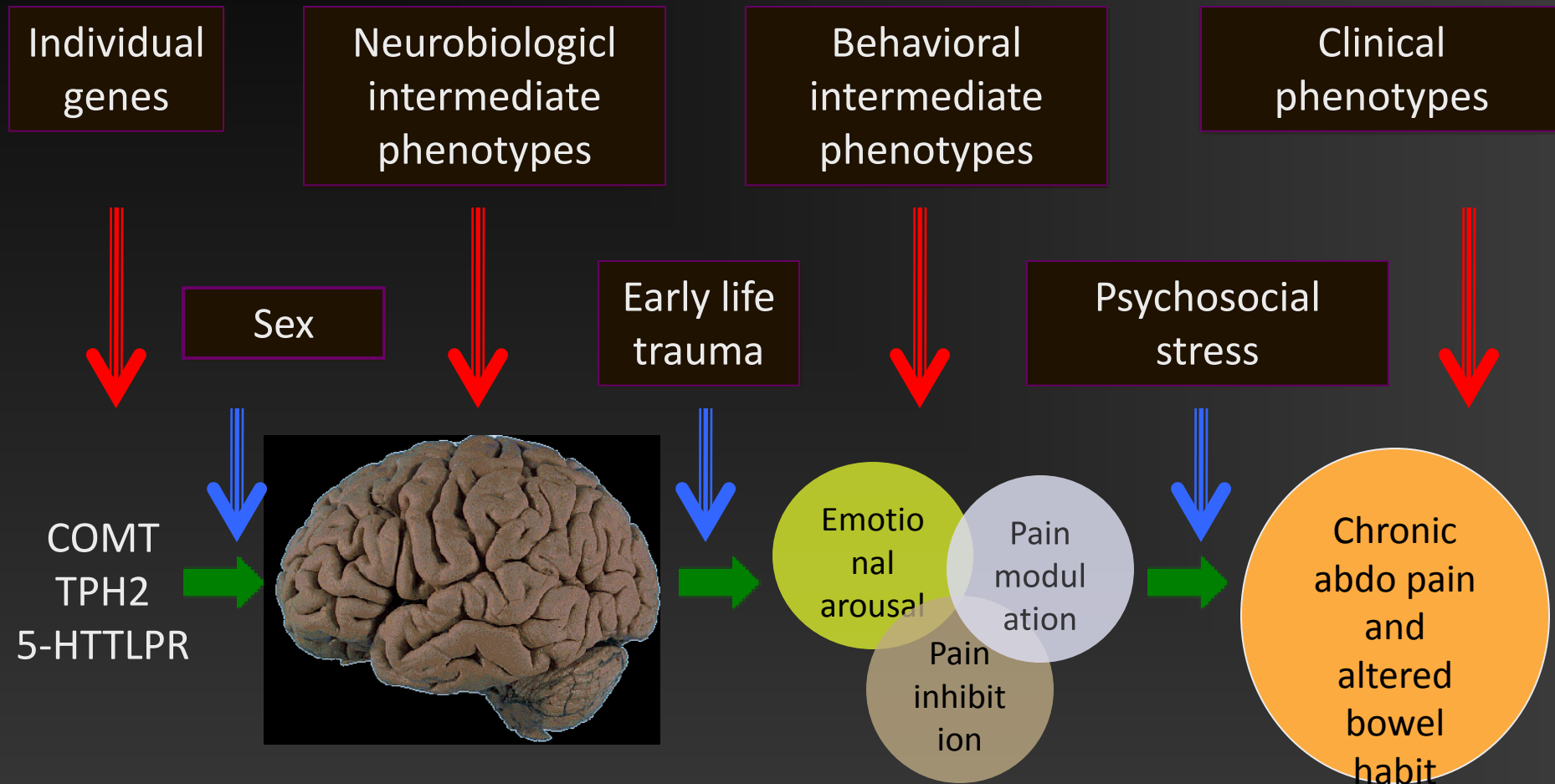


Somatic / functional pain complaints

Explanations

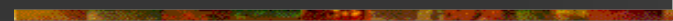
- ☀ Control of pain processing in insula related to a number of Genes
- ☀ Various events can alter gene expression :
 - ☀ Major stress/trauma
 - ☀ Infection
 - ☀ Physical/sexual abuse
- ☀ Vulnerable genotypes susceptible to such events leading to changes to central processing in insula.
- ☀ Altered autonomic and control functions change and end organ function

Explaining function disorders: the endophenotype approach to IBS

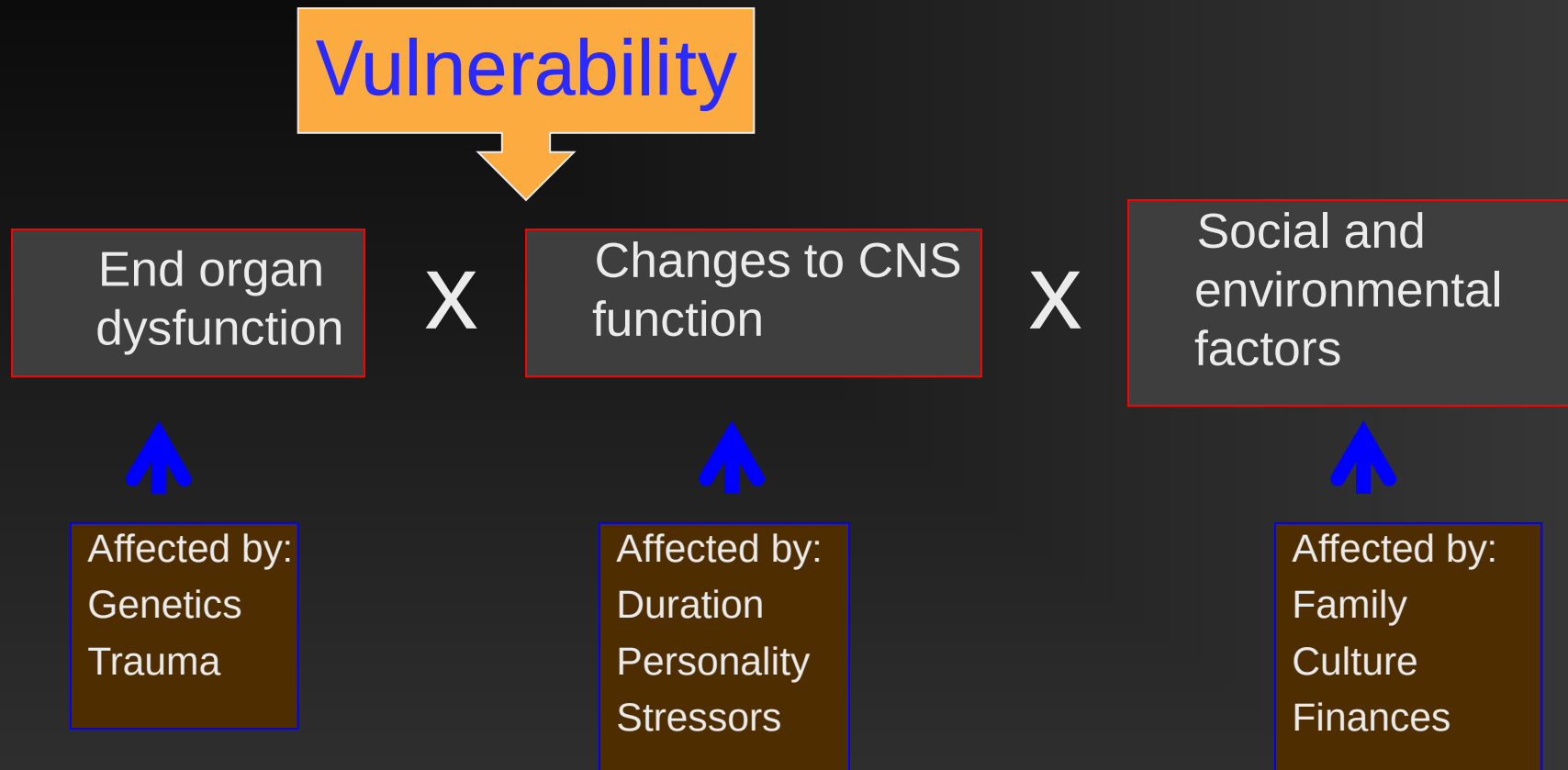


Mayer 2009

- Pain perception involves the activation of a complex matrix of cortical activity
- Aspects of this matrix can be activated by attributions, beliefs, fears without tissue damage
- Past experience and learning contribute to behaviour
- Cultural factors dictate the response to suffering

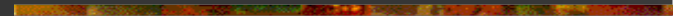


Conclusions



Chronic pain syndrome(s)

- ✱ Chronic pain with disability and distress
- ✱ Frequent history of other troublesome somatic symptoms
- ✱ Does not imply formal 'syndrome' (e.g. Laurence Moon Beedle)
- ✱ Not recognised in IASP definitions of pain states
- ✱ Has been described and can be referenced
- ✱ Not a psychiatric DSM IV diagnosis (closest somatoform pain disorder)
- ✱ Relevant conditions: IBS, FMS, ME, LBP, etc



Low back Pain

The problem of back pain

About 50% of the population will experience back pain in any year.

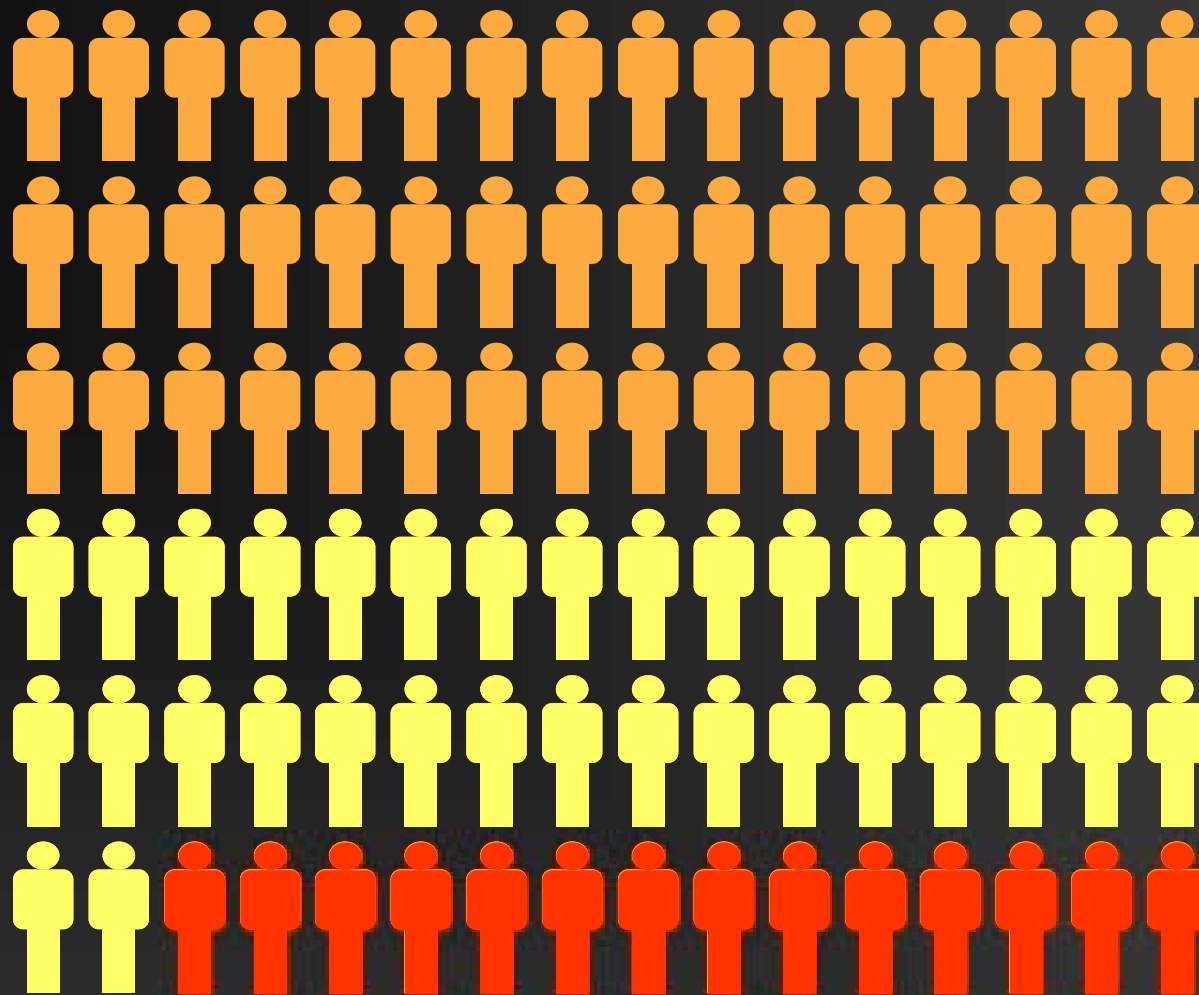


6-9% of the population will consult their GP on account of it

Polyclinic (50,000) this equates to 4,000 GP consultations per annum

Of the people
who consult
their GP 37%
will rate their
pain as very
bothersome....

and 14% as
extremely
bothersome



In a polyclinic this equates to 500 + people

Bothersomeness

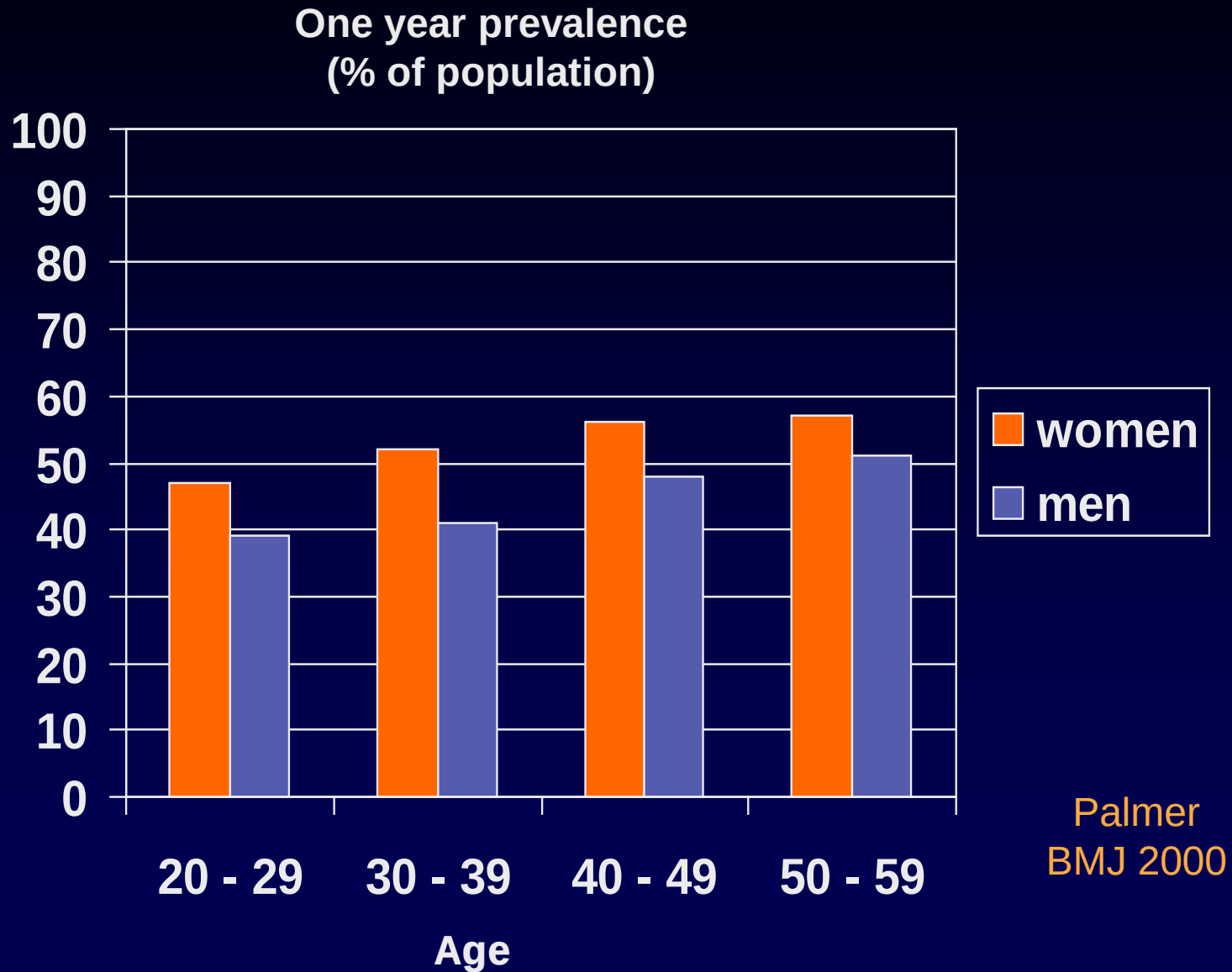
- ☀ 1464 patients consulted with back pain (9.3% of pop)
- ☀ Analysis of 929 follow ups at 6 months

	Not at all	Slightly	Moderately	Very much	Extremely
%	6	13	30	37	14
Pain VAS	0.3	2	4.4	6.5	7.7
R & M	1.5	3	7.5	12.8	15.7
>11 HAD anx	15.7	10.8	28.1	45.4	53.5
>11 HAD dep	3.8	3.3	13.9	31.1	50.7
Not working %	0	3.2	13.1	51.1	51.3

About 10% of those off with pain at a month will be off work

Dunn & Croft Spine 2005

One year prevalence of back pain



Prediction of chronicity & disability

The development of chronicity following an episode of musculoskeletal pain is predicted by psychosocial factors not physical or biological

Prognostic risk factors for poor outcome:

- Pain severity
- Anxiety/depression
- High somatic perception
- Adverse coping
- High disability
- Reduced ROM
- Longer duration
- Multiple sites
- Previous episodes
- Older age
- Low social support

Measure of disability, QOL etc

days

weeks

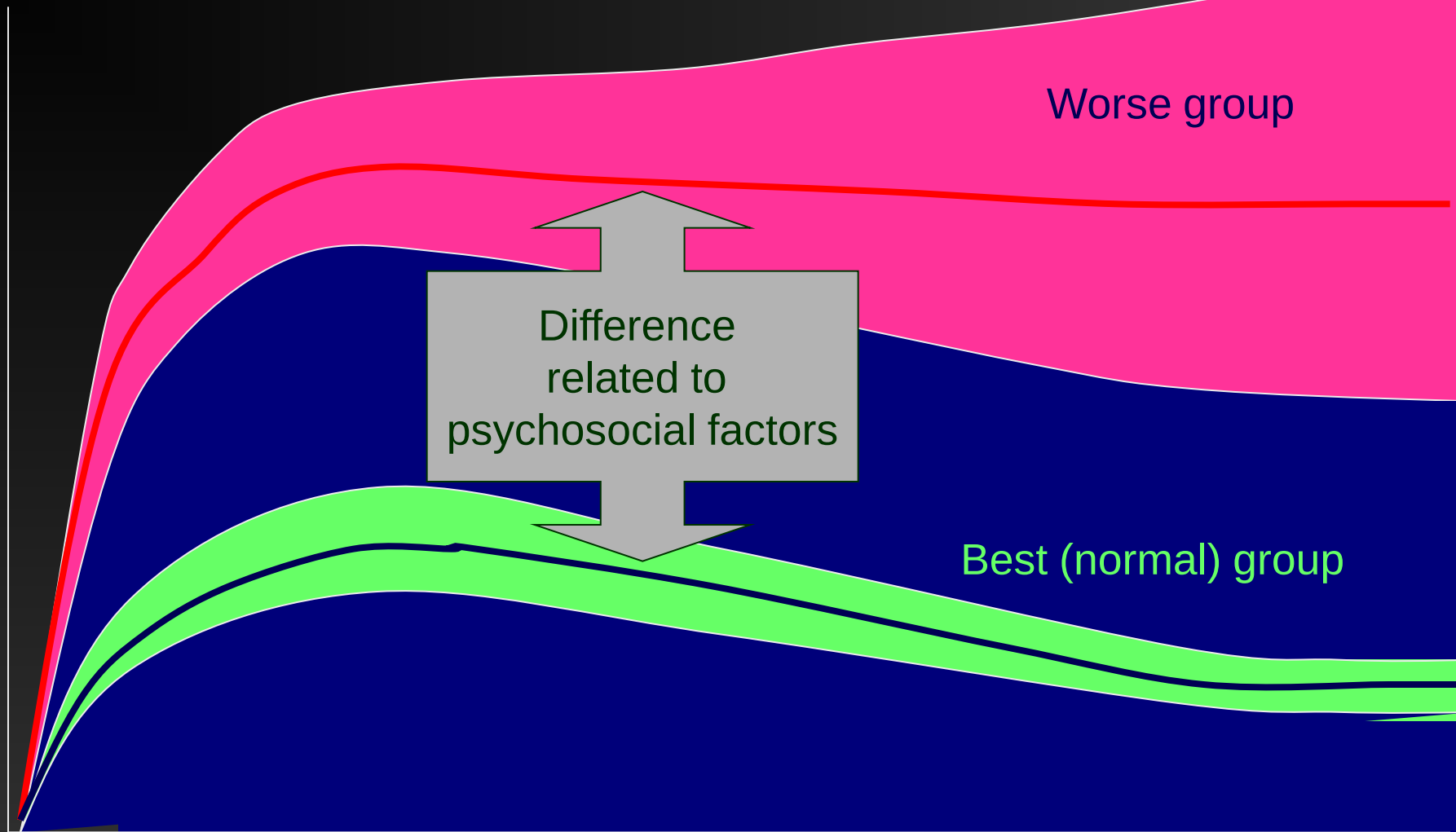
months

years

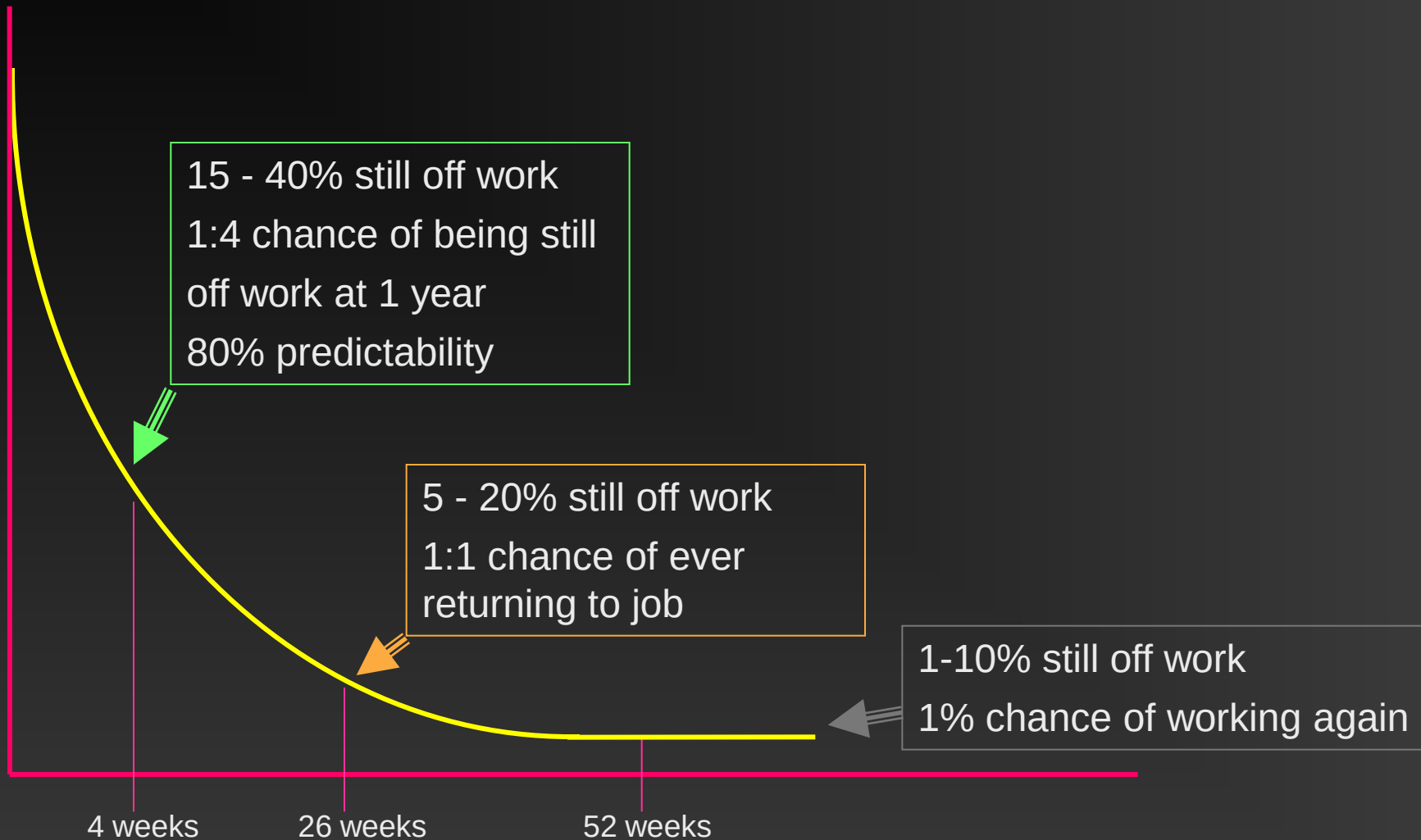
Worse group

Difference
related to
psychosocial factors

Best (normal) group



Return to work



Doctors are just the same as lawyers; the only difference is that lawyers merely rob you, whereas doctors rob you and kill you as well.

Anton Chekov



The philosophies of one age
have become the absurdities of
the next, and the foolishness of
yesterday has become the
wisdom of tomorrow.

William Osler

'Patients suffer illnesses; doctors diagnose and treat diseases.

Traditional healers redefine illness as disease: because they share symbols and metaphors consonant with lay beliefs, their healing rituals are more responsive to the psychosocial context of illness

When physicians dismiss illness because ascertainable disease is absent they fail to meet their socially assigned responsibility. It is essential to reintegrate scientific and social concepts of disease and illness as a basis for a functional system of medical research and care.'

Leon Eisenberg 1977

Back pain, a communicable disease?

Background

Back pain (BP) is a frequent disorder affecting currently up to 40% of adults in Western Europe. Most of it is said to be 'non-specific', i.e. lacking an obvious patho-anatomical explanation..

Aim

To propose the hypothesis of BP being a communicable disease.

Results

We base our hypothesis on a reanalysis of five German health surveys. They show a wide gap in BP prevalence between West and East Germany early after reunification. The gap consistently decreased to nearly zero in 2003. Work disability data followed a comparable course.

Prevalence of whiplash symptoms following placebo collision

Castro et al 2001

- ✱ Complex protocol involving simulated rear end collision
- ✱ 51 subjects, history and physical examination taken at T0, T1 just after, T2 3 days after, T4 4 weeks after.
- ✱ Various psychological measures taken alongside symptom analysis
- ✱ Symptoms (fright, pain, dizziness, headache, nausea, tinnitus, fatigue, vomiting) initially in 20% reduced to 10% by T4
- ✱ A group of psychological factors predicted 83-92% development of symptoms
- ✱ Emotionally unstable, less happy with life and more concerned with health more at risk



“When a lot of remedies are suggested for a disease, that means it can't be cured.”

Anton Chekov

“ It is incident to physicians,
I am afraid, beyond all other
men, to mistake
subsequence for
consequence.”

Samuel Johnson
(1709–1784).

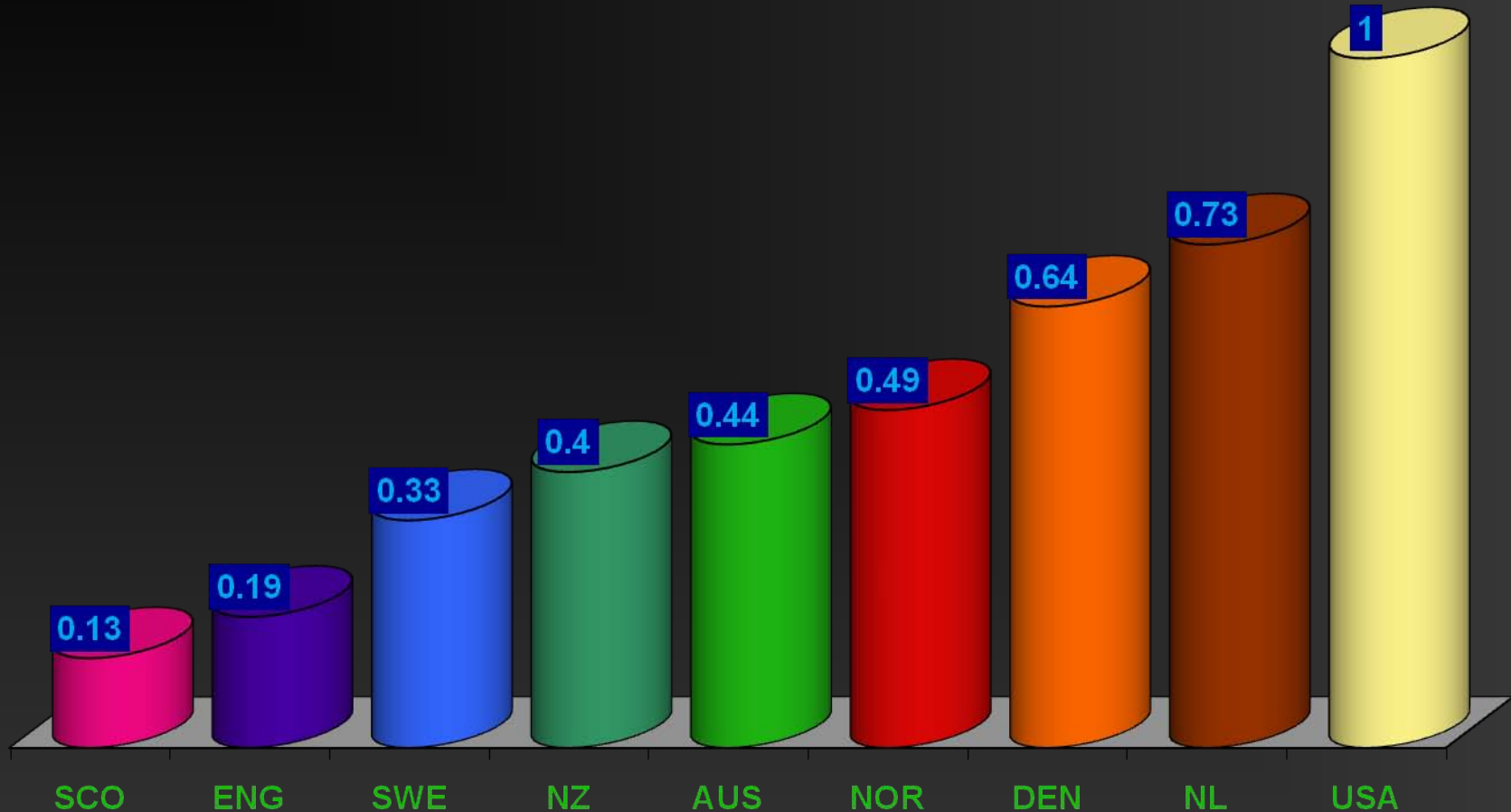


It is impossible for the doctor treating a patient with a chronic condition to know whether the treatment he or she is providing is effective.

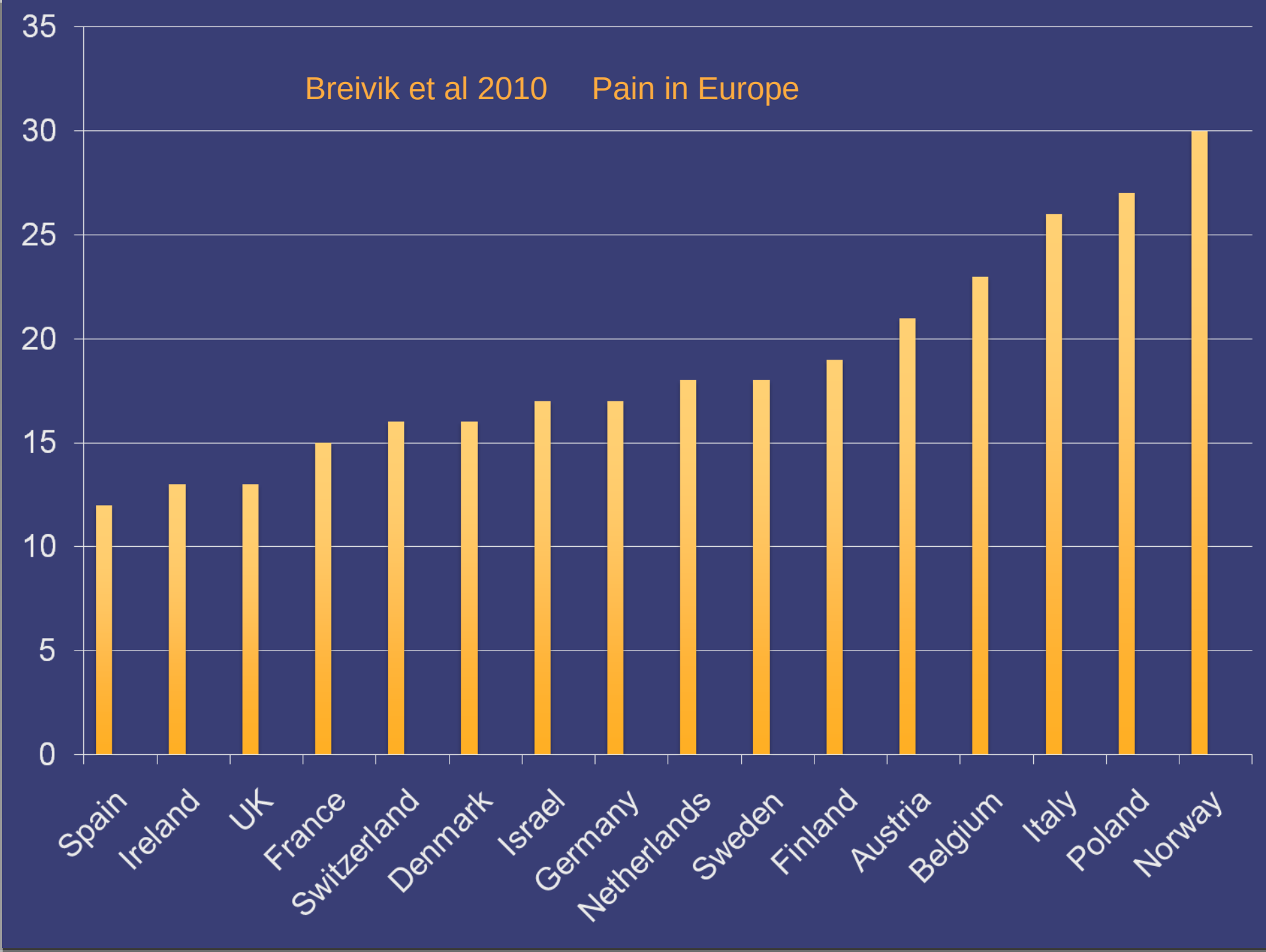
Because:

- Conditions wax and wane
- Timing of consultation when symptoms bad
- Patients are nice to their doctors
- Misattribution is common (by both)
- Placebo and non specific effects

Global spinal surgery rates (standardised to US = 1)



Breivik et al 2010 Pain in Europe



Are the interests of the individual the same as those of the herd?



What does society want?

- More productivity
- Less disability
- Less resource utilisation
- Less costs

‘Culture makes pain tolerable by integrating it into a meaningful setting; cosmopolitan civilisation detaches pain from any subjective or inter-subjective context in order to annihilate it. Culture makes pain tolerable by interpreting its necessity; only pain perceived as curable is intolerable.’ I Illich

Google/Medline index

- ☀ <1 Healthy
- ☀ >1 Topical
- ☀ >10 Concern

☀ Disc replacement	4,220,000 / 739	5,710
☀ Facet joint injection	115,000 / 274	420
☀ Fibromyalgia	8,170,000 / 6076	1,344

Chronic pain can get better

Nuprin pain report 1986 18%

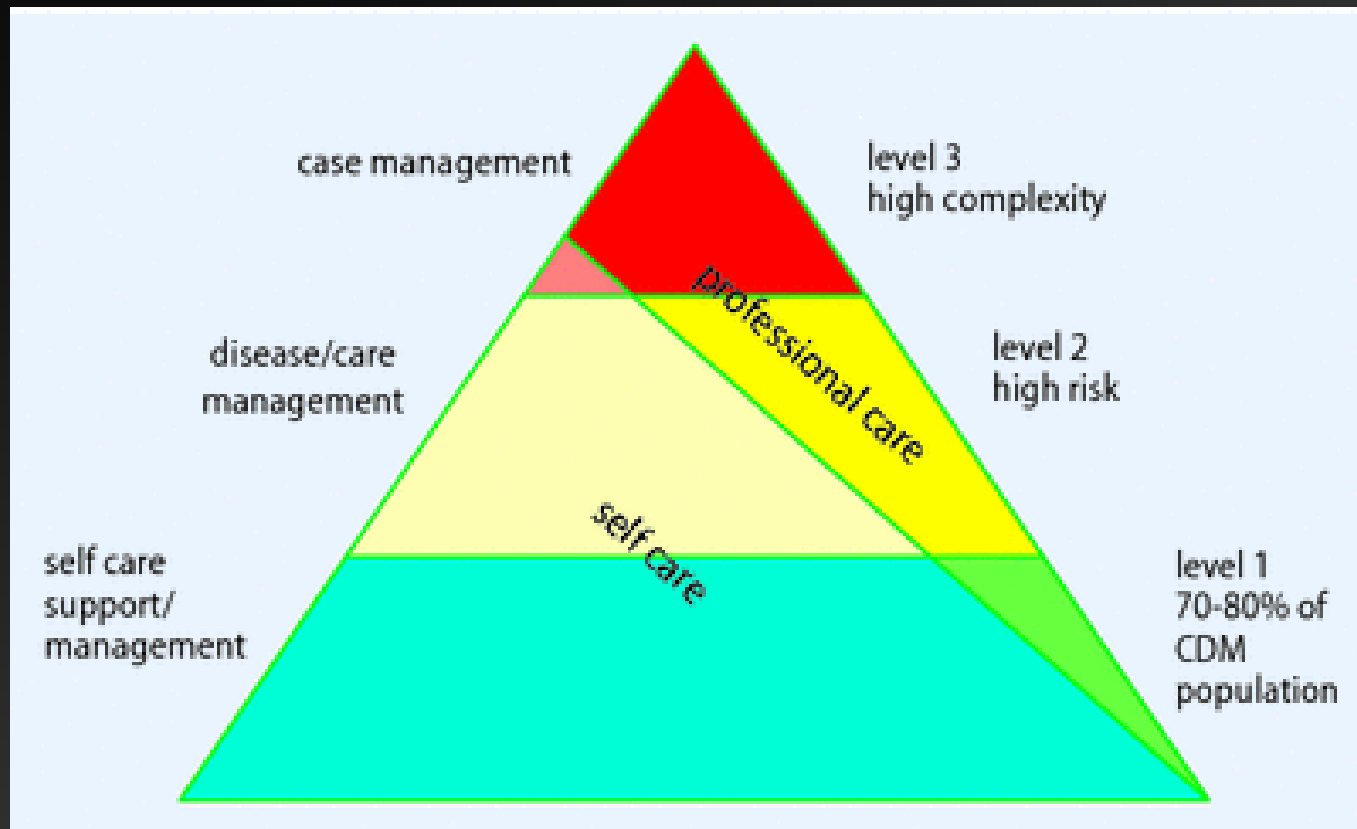
Blyth Pain 2001 17-20%

Breivik 2010 19%

"There is far less knowledge about the fact, that about 90 percent of pain is self-limiting that it is not always an indication of poor health, and that, most frequently, it is the result of tension, stress, worry, idleness, boredom, frustration, suppressed rage, insufficient sleep, overeating, poorly balanced diet, smoking, excessive drinking, inadequate exercise, stale air, or any other abuses encountered by the human body in modern society"

Norman Cousins, *Anatomy of an illness*,
Page 90, "Pain is not the enemy"

Chronic illness management



Leeds

Fawlty Towers

Railtrack

The psyche

What is a schema ?

- ✦ a schema is a psychological concept, which is
 - ✦ a structure or framework
 - ✦ comprising a cluster of facts, ideas, behaviours and emotions
 - ✦ that enable us to understand the world and
 - ✦ transfer appropriate knowledge and skill to new situations

Schema in pain management

- ✱ Our patients have schema associated with their pain and the experience of their illness
- ✱ This will be a hugely complex matrix of 'knowledge', beliefs, and fears, fuelled by folk myth, family experience and lay and professional medical advice
- ✱ The journey will have been modified by physical and emotional hurt, encounters with treatment and expectations.
- ✱ To enter the schema is to enter its emotional space
- ✱ To change such schemata requires intensive, consistent cognitive and behavioural treatment

But remember doctors have schema too....!

Schema in pain management

- Schemata influence our attention, as we are more likely to notice things that fit into our schema.
- If something contradicts our schema, it may be encoded or interpreted as an exception or as unique.
- Schemata are prone to distortion. They influence what we look for in a situation. They have a tendency to remain unchanged, even in the face of contradictory information.
- We are inclined to place people who do not fit our schema in a "special" or "different" category, rather than to consider the possibility that our schema may be faulty.
- As a result of schemata, we act in such a way that actually causes our expectations to come true”.

Schema in pain management

When we help a patient we change
their schema...

But when we change their schema we
help the patient..

Effective pain management

What should we be striving for?

☀ Pain relief?

- ☀ • Don't kill me
- ☀ • Don't hurt me
- ☀ • Don't make me feel helpless
- ☀ • Don't keep me waiting
- ☀ • Don't waste resources

☀ • Donald Berwick Annals Int Med 2005

Effective pain management

Aims of pain management:

- ✱ Ease the pain, but never let it be the sole goal
- ✱ Change the schema (understanding, thoughts, beliefs, emotions)
- ✱ Change the behaviour (activity level, fitness, sleep, health seeking, work)
- ✱ Translate into the home environment

Effective pain management

CBT programmes

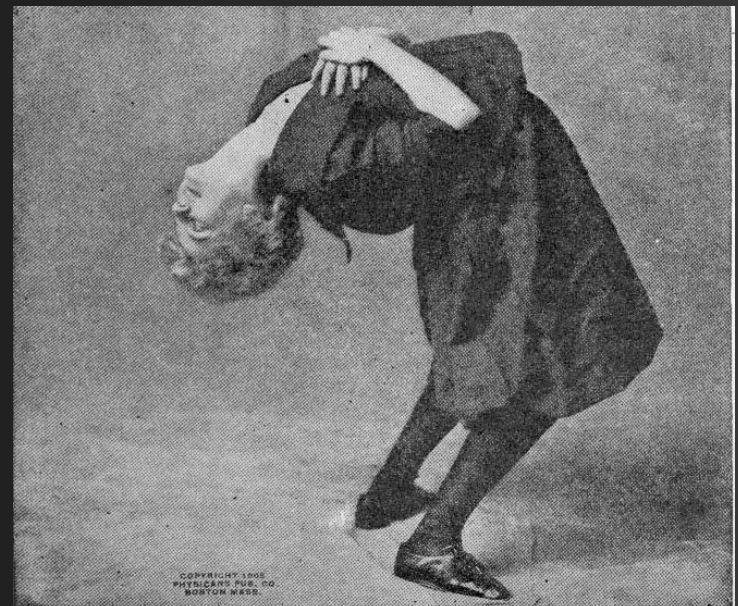
- ✱ Its not about 'coping' with the pain
- ✱ Behaviour change is king
- ✱ Pain is not a life sentence
- ✱ Pacing is about increasing activity
- ✱ The group does as much as the staff
- ✱ Intense treatment works better
- ✱ Doctors should be involved

HRG4	Treatment Details	OPCS
AB01Z	RF Spinal Nerve Root	A573
AB02Z	RF Specific Nerve	A604
	RF Cervical Sympathetic Nerve	A781
	RF Lumbar Sympathetic Nerve	A783
	RF Facet Joint Cervical	V481
	RF Facet Joint Thoracic	V483
	RF Facet Joint Lumbar	V485
AB03Z	Facet Joint Injection	V544
	Stellate Ganglion Block	A811
AB04Z	Sympathectomy	A812
	Sacro-iliac Joint Injection	X308
	Lumbar Epidural	A521
	Sacral Epidural	A522
	Cervical/Thoracic Epidural	A528
	Unspecified Epidural	A529
	Guanethidine Block	X298
	Other Specified Injection of therapeutic substance	X308
AB06Z	Acupuncture	A706
	MDT CBT based Pain Management Programme	X662
	Uni-disciplinary CBT patient education	X661
	CBT (other specified)	X668
	CBT (unspecified)	X669
	TENS	A707
	Manipulation	X558
	Trigger Point Injection	X306
	Other Joint Injection	W904

Physiotherapy and exercise

“ take the following exercises twice a daybefore exercising take a tepid bath made brisk with aromatic vinegar.....
the exercises consist of walking or running around the room on all fours and bending backwards until the hands almost touch the floor....”

Virtues Household Physician



A severe exercise for strengthening the back.



Good exercise for stout women.

Systematic review of exercise therapy for low back pain

van Tulder et al Spine 2000;25;21;2784

- ✱ 39 papers since 1990 review
- ✱ strong evidence that exercise is not more effective for acute back pain than other inactive or active treatments
- ✱ conflicting evidence compared with inactive treatments for chronic pain
- ✱ exercise better than usual GP care and as good as conventional physio
- ✱ exercise probably aided resumption of normal activity

Acupuncture for back pain: a systematic review and meta-analysis

- ✱ 70 RCTs identified, 45 excluded leaving 25 of sufficient quality
- ✱ Grouped according to: acute / chronic, style of acup, controls
- ✱ Results
 - Short term – more effective than sham or no treat
 - Significant improvements in long & short term pain relief, and overall improvement
 - Little change in function, analgesic use or return to work

Manheimer et al 2004



Epidural injection



- ✱ Many studies most of poor quality
- ✱ At least three systematic reviews (e.g. Koes 1995, Nelemans 2001)
- ✱ Much anecdotal evidence
- ✱ Synthesis:
 - ✱ Limited supportive evidence
 - ✱ Greater efficacy for leg pain than for back pain
 - ✱ No long term gains
 - ✱ Steroid probably contributes
 - ✱ Site of injection (caudal vs lumbar) not important
 - ✱ Ineffective for chronic back pain

Injection studies in spinal disorders

Centre for Spinal Surgery, University of Zurich, Zurich, Switzerland.



“...Despite the widespread use of these spinal injections, their application is based widely on anecdotal experience and evidence that is not evidence based.

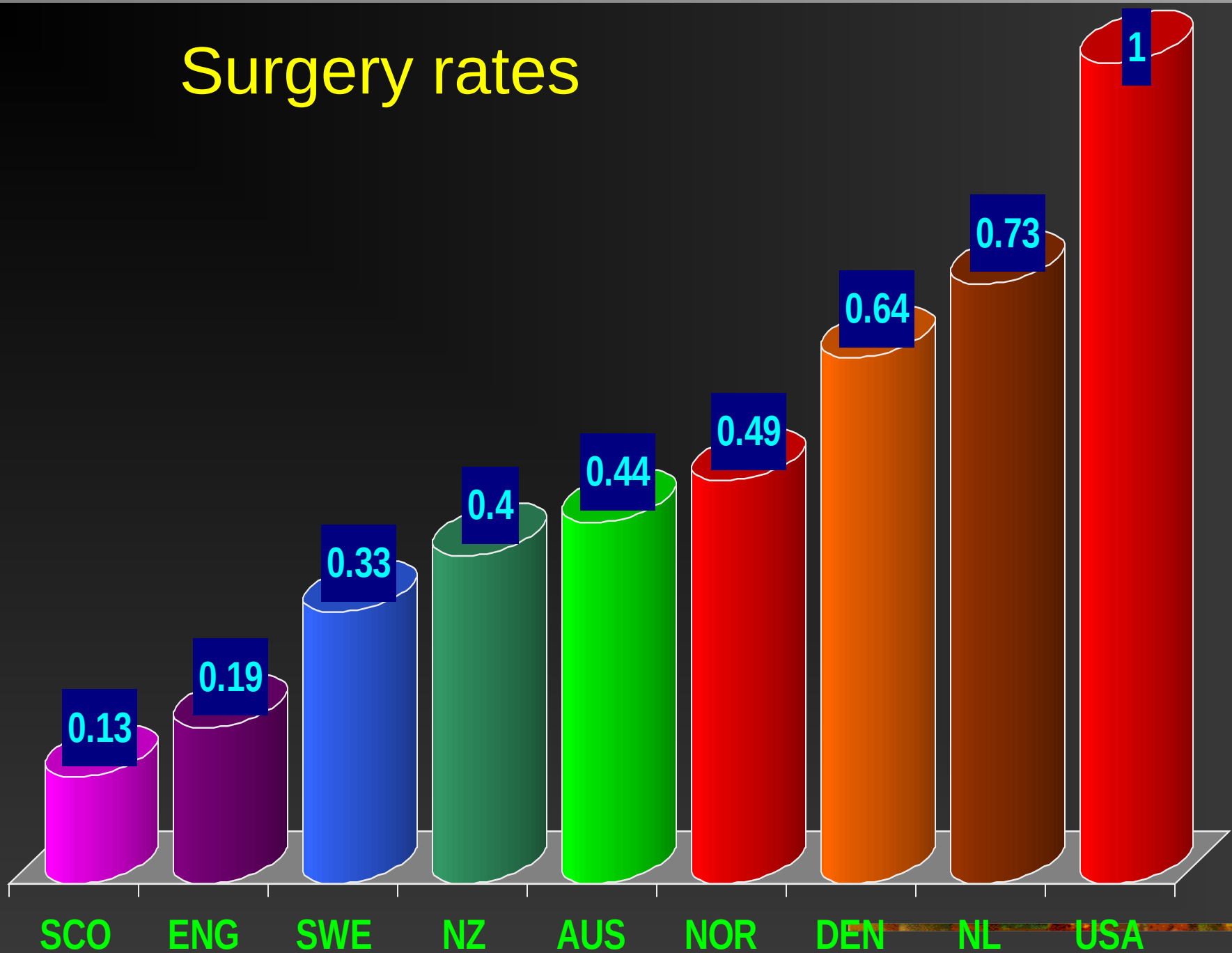
We highlight the need for prospective randomized trials on the clinical efficiency of spinal injections and stress the importance for future considerations of cost effectiveness to justify their further application”.

Loenardi, Pfirrmann & Boos, Clin Orthop Relate Res Feb 2006

Spinal cord stimulation for chronic back and leg pain and failed back surgery syndrome: a systematic review and analysis of prognostic factors.

- ✱ Overall, 43% of patients with chronic back and leg pain/failed back surgery syndrome experienced one or more complications following a spinal cord stimulation implant, although no major adverse events were reported.
- ✱ Despite an increase in the number of studies over the last 10 years, the level of evidence for the efficacy of spinal cord stimulation in chronic back and leg pain/failed back surgery syndrome remains "moderate."
- ✱ Prognostic factors found to be predictive of the level of pain relief following spinal cord stimulation were study quality, follow-up duration, study setting, and patient indication.

Surgery rates



Randomised controlled trial of surgical stabilisation of the lumbar spine with an intensive rehabilitation programme for patients with chronic low back pain.



-Spinal fusion surgery has only modest, if any, effects

Conclusion

- No clear evidence that surgery is better than an intensive rehabilitation programme

Fairbank et al BMJ 2005

Spinal surgery vs functional restoration

- ☀ 349 patients, pain > 1 year
- ☀ multicentre, randomised to spinal fusion or outpatient functional restoration programme
- ☀ No difference in outcome at two years except in Oswestry disability score

surgery	46.5	34.2
FRP	44.8	35.7

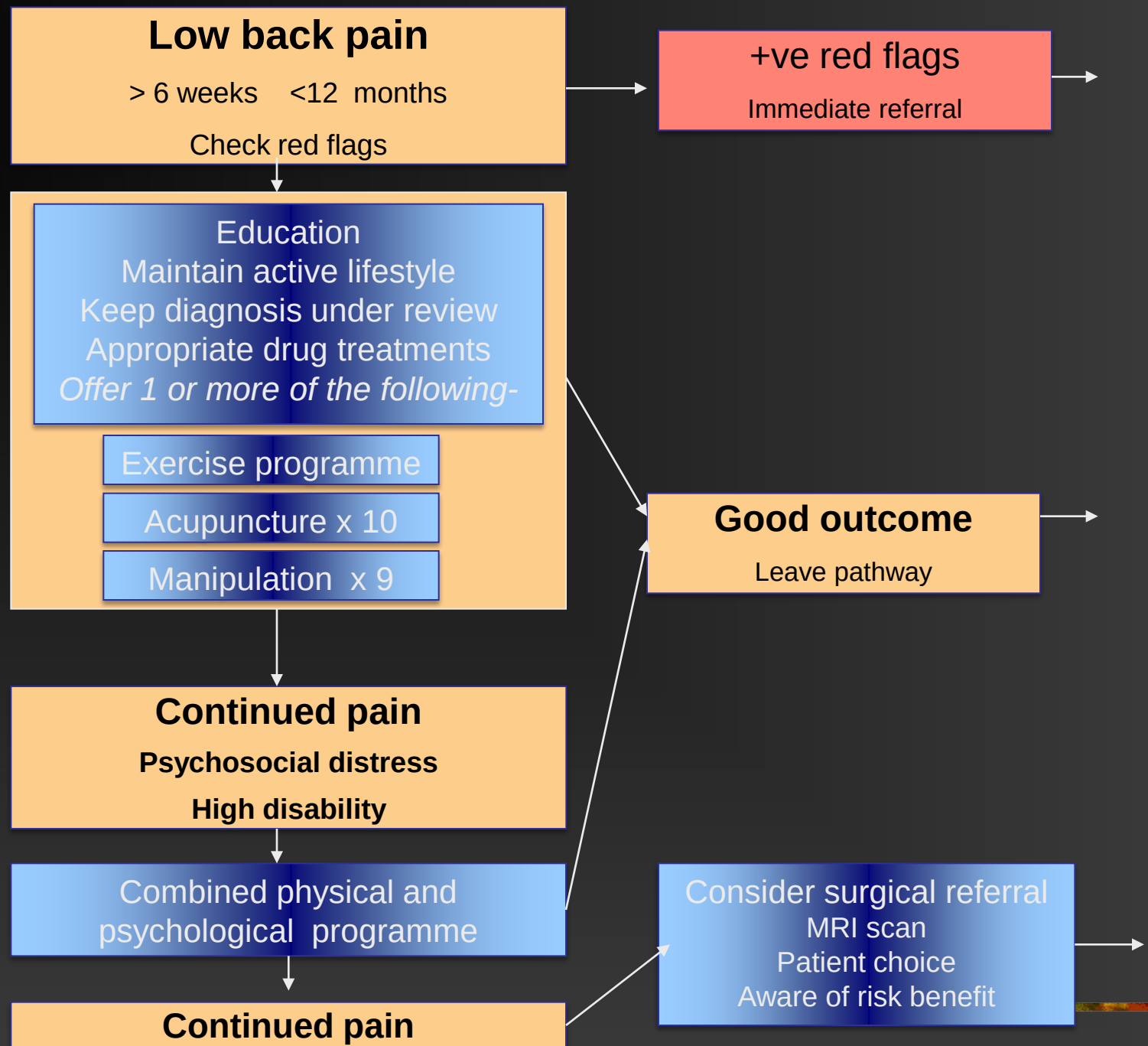
☀ Conclusion

- ☀ No clear evidence that surgery is better than an intensive rehabilitation programme

NICE
2009

Back Pain

6 weeks
to 1 year



Recommendations of the American College of Physicians and American Pain Society for management of back pain

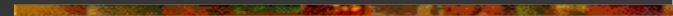
Recommendation 1: Clinicians should use history & examination (including psychosocial factors) to place patients into 1 of 3 categories: nonspecific low back pain, back pain associated with radiculopathy or spinal stenosis, or back pain potentially associated with another specific spinal cause.

Recommendation 2: Clinicians should not routinely obtain imaging or other diagnostic tests in patients with nonspecific low back pain

Recommendation 3: Clinicians should perform imaging for patients when pain severe or with progressive neurologic deficits or when serious underlying conditions are suspected.

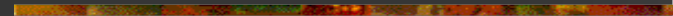
Recommendation 4: Clinicians should evaluate patients with persistent pain and signs or symptoms of radiculopathy or spinal stenosis with MRI or CT only if they are potential candidates for surgery or epidural injection

Multidisciplinary treatment



What do we mean by multidisciplinary?

- ✱ Different groups of doctors?
 - ✱ Neurologist
 - ✱ Rheumatologist
 - ✱ Anesthesiologist
- ✱ Doctors working alongside other clinical staff?
 - ✱ Doctor
 - ✱ Nurse
 - ✱ Pharmacist
- ✱ A clinical team that may include a doctor?
 - ✱ Nurse
 - ✱ Physiotherapist
 - ✱ Occupational therapist
- ✱ Any old team
 - ✱ Tai Chi teacher
 - ✱ Knitting therapist
 - ✱ Tea lady



What do we really mean by multidisciplinary?

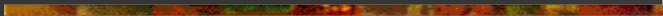
- ✦ A multidisciplinary team comprising at least
 - ✦ Psychology
 - ✦ Physiotherapy
 - ✦ medicine
 - ✦ Working on a broader model than the biological
 - ✦ With at least some attempt at addressing psychological aspects of the persons condition such as
 - ✦ Understanding
 - ✦ Fears
 - ✦ Beliefs
- along with physical reconditioning

Implication: biopsychosocial



Assessment

After Turk 1993



Yellow flags

- ✱ Belief back pain is harmful or potentially severely disabling
 - ✱ Fear avoidance behaviour or reduced activity level
 - ✱ Tendency to low mood and social withdrawal
 - ✱ Past history of stress and depression
 - ✱ Expectation that passive treatment rather than active participation will help
 - ✱ Poor sickness absence record
 - ✱ Poor interaction with work colleagues
- 

What is CBT?

- ✱ Actions (behaviours) are preceded by thoughts
 - ✱ Thoughts are informed by knowledge and beliefs
 - ✱ Feelings (emotions) are predicated by thoughts
 - ✱ Patterns of behaviour whilst driven by beliefs can reinforce them
 - ✱ We may not seem to have a choice of feelings but we can check and challenge our thoughts
 - ✱ Changing behaviour requires an understanding of these processes and examining assumptions and fears
 - ✱ CBT is collaborative and negotiated putting the subject first rather than simply educational or directive
-

Multidisciplinary bio-psycho-social rehabilitation for chronic low back pain

- ✱ Systematic review, 10 trials 1964 randomised patients
- ✱ Strong evidence that intensive treatment improved function when compared with inpatient or outpatient non interdisciplinary treatment and reduces pain.
- ✱ Less intensive treatment (<100hrs) were less effective

(Guzman 2004)

Efficacy of behavioural approaches in low back pain

van Tulder et al Spine 2000;25;2688

- ✱ 21 RCT's of behavioural treatment divided into cognitive, operant and respondent
- ✱ 6 studies of high quality
- ✱ strong evidence that BT has a moderate positive effect on pain intensity, general function and behavioural outcomes when compared to waiting list or no treatment
- ✱ mod evidence of no short term effect of adding behavioural treatment on pain or functional status to treatment as usual for chronic low back pain

Treatment components

- ☀ physical reactivation / rehabilitation

- ☀ strength,
- ☀ range of motion
- ☀ Fitness

PT

- ☀ psychological therapy

- ☀ CBT

Psych

- ☀ education

All

OT

Nurse

OT/Psych

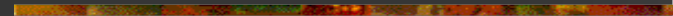
- ☀ pacing, planning & goal setting

- ☀ drug reduction

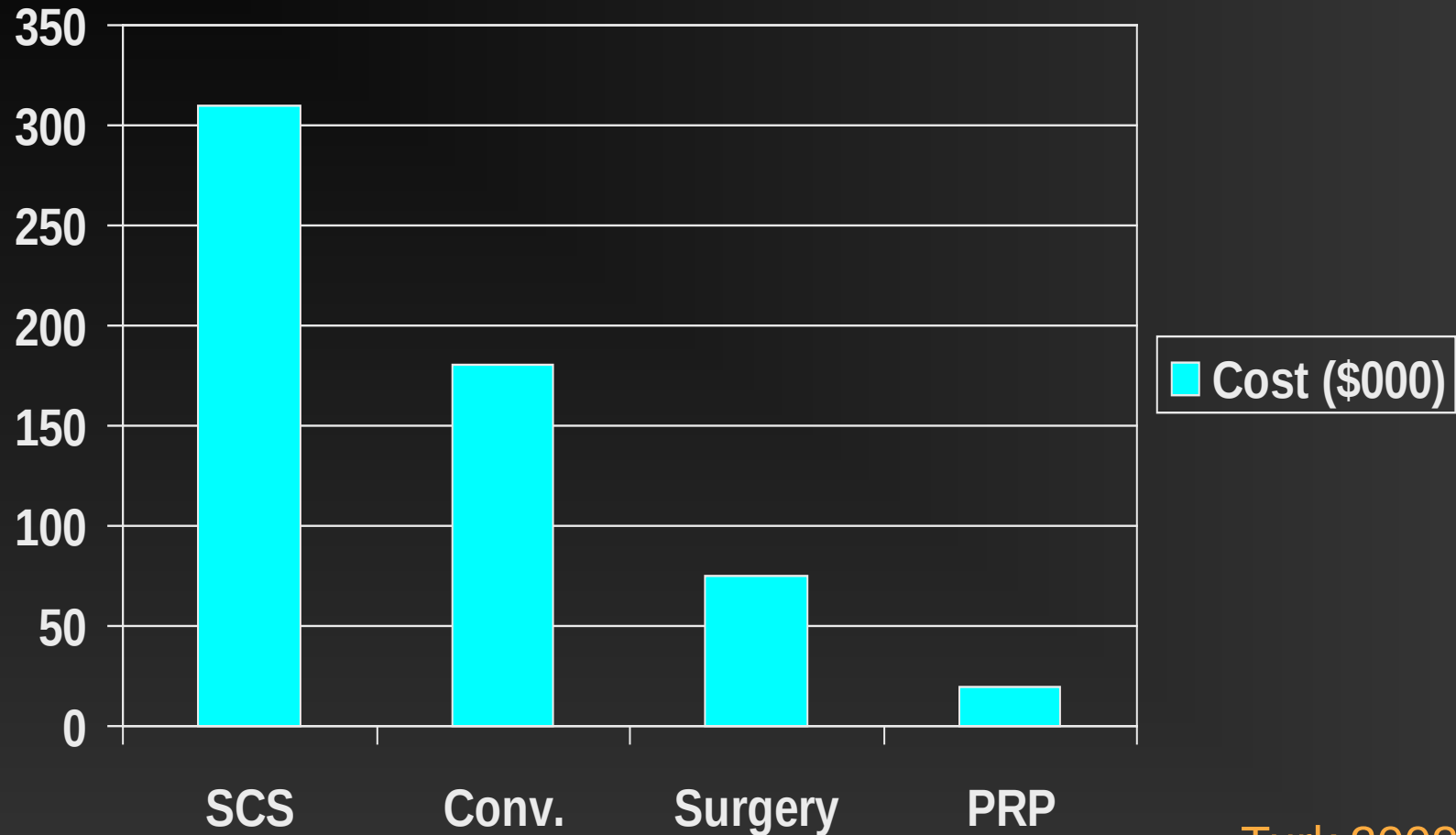
- ☀ return to work

What are we trying to do on a PMP?

- ✱ improve knowledge about chronic pain
- ✱ Instil notion of self management
- ✱ cognitive training/stress management
- ✱ exposure to feared postures, tasks and activities
- ✱ graded exercise programme
- ✱ pacing up sitting, standing, walking
- ✱ reduce use of sticks, collar, corset etc
- ✱ teaching relaxation
- ✱ reduce unhelpful medications



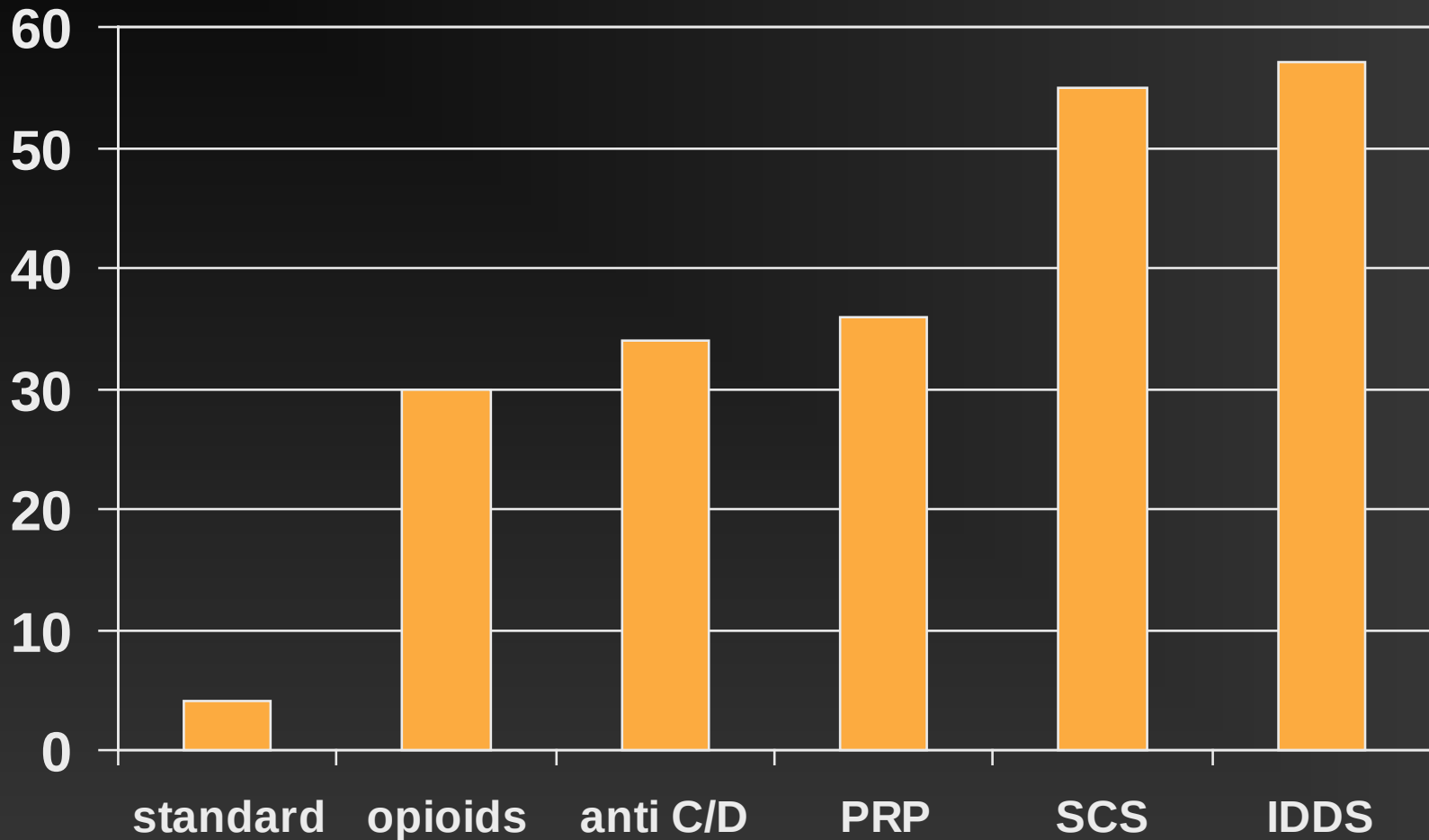
Cost to return to work one patient



Turk 2002

Percentage pain reduction after various pain treatments

Turk 2002



The *RealHealth* Institute

The RHI is an ethical Healthcare company offering high quality evidenced base interdisciplinary treatment for musculoskeletal complaints, pain, stress and fatigue

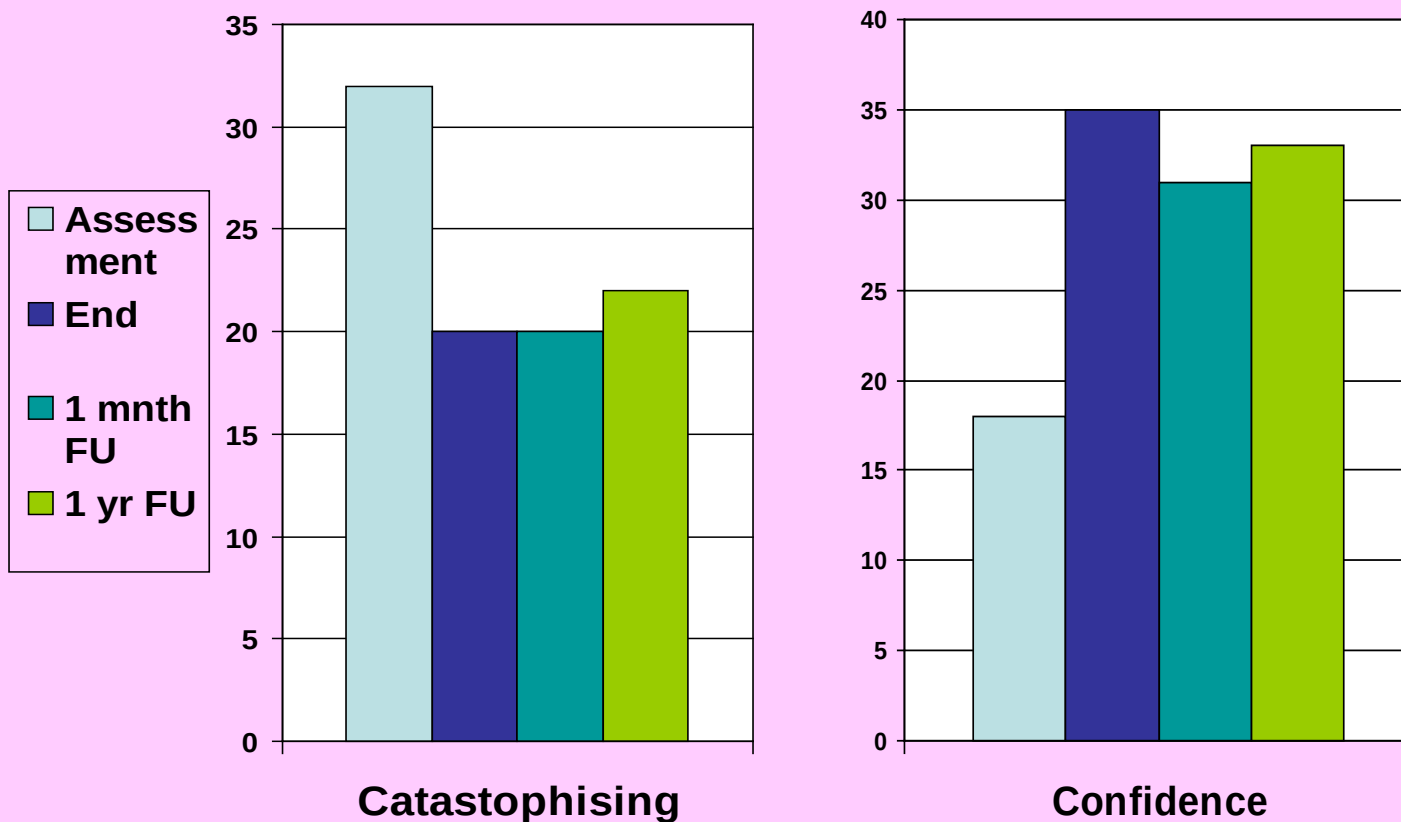
Services include

- ☀ Triage and assessment
- ☀ Interdisciplinary treatment programmes
- ☀ Individualised treatment packages
- ☀ Consultancy and advice
- ☀ Training and education



Results from 4 week pain management programme

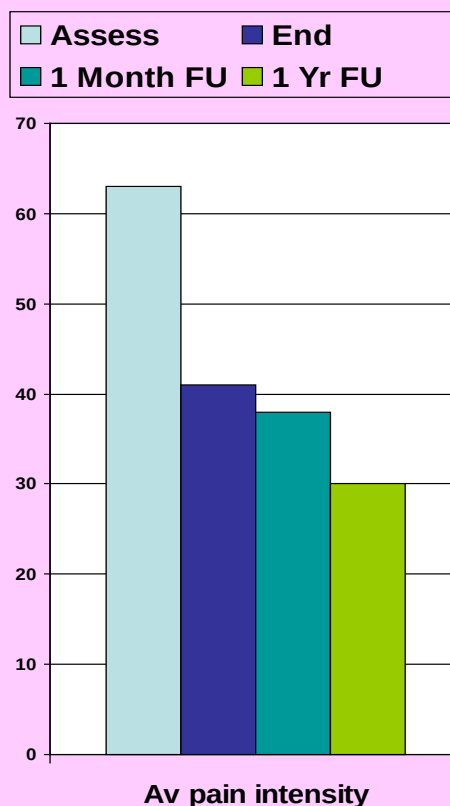
Substantial improvements in confidence and depression with a reduction in catastrophising and improvements in physical function



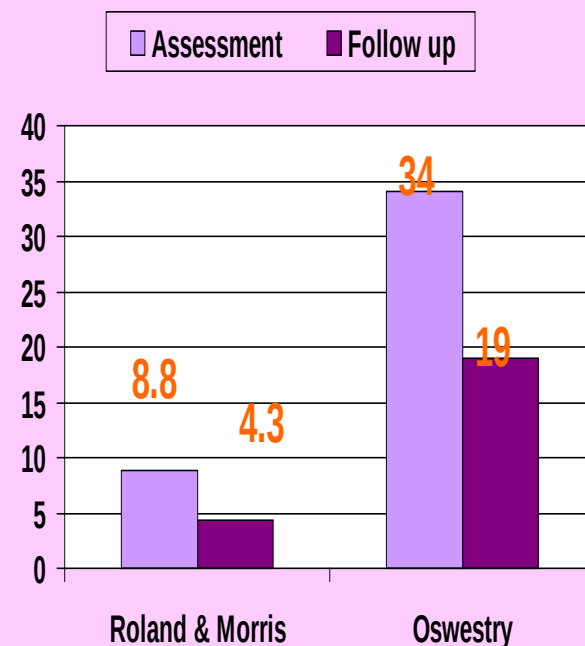
Results from 11 day Functional Restoration Programme

- 135 patients (57male 78 female)
- 3 week 75 hour FRP
- 26 month follow up
- Scandinavian Global outcomes
 - 85% better or much better
 - 84% excellent or good

Reduction in pain



Halving of disability scores



I, not events, have the power to make me happy or unhappy today. I can choose which it shall be. Yesterday is dead, tomorrow hasn't arrived yet. I have just one day, today, and I'm going to be happy in it.

Groucho Marx

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